COS30049 Segment: Internet of Things (IoT)

A 6-Week Crash-Course on IoT in 2 Hours

beginning with

Project Proposal & Architecture Design

Prepared by Dr. Mark Tee Kit Tsun (SUTS)

Checklist

1. Select an IoT Person-In-Charge from your team – Sign and Submit **Equipment Loan Form**
2. Collect IoT Auxiliary Kit (1 per team) – Must be returned in Week 12
3. Content Check – Guided
4. Doublecheck access to Major Specific Content – IoT
5. Onwards with the Overview

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An IoT (or general Software) Proposal



- There is no single standard Form to fill (unless the client has one).

- However, every Project Proposal should have these components:
  a) **Background / Problem Description**
     - Describes the client company, operating environment, what problems are faced, and why the clients need a solution.
     - Written from your (the Development Team's) point of view.
     - Shows your understanding of the Client's issues.

  b) **Scope Statement**
     - What is the aim and capability of the proposed solution.
     - Why can it solve the client's problem(s).
     - Usually consist of:
       - Goals – the general purpose of the solution.
       - Objectives – specific attributes used to measure the success of the project (must be S.M.A.R.T.).
       - Constraints – Conditions and Limits that must be adhered to.
       - Out of Scope – Specific things that the Team will not be working on.

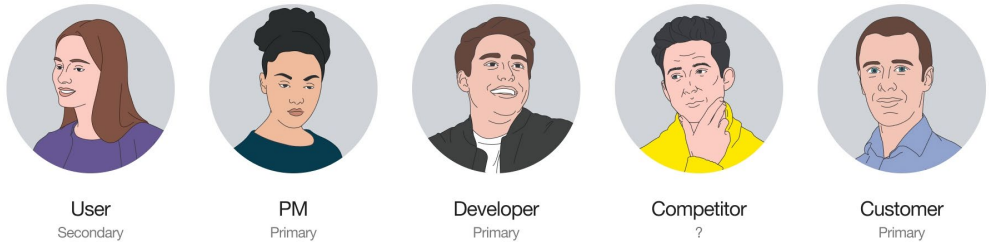


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An IoT (or general Software) Proposal

c) Stakeholders

- Who is involved in the project.
- Can include Project Sponsor, Customer/Client, Developers, and/or more.
- Describes the Users who interact with the system.



d) Deliverables and Schedule

- Indicate what are the items to deliver, and when (e.g., Installers, Servers, Nodes, Manual, etc.)
- The Timeline:
  - use calendar dates and Sprint Numbers to indicate when the expected completion and submission of deliverables is.
  - Use a Gantt Chart to visualize it.
  - Consider things like Dependencies, Risks involved and Business Value.



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An IoT (or general Software) Proposal

e) Solution Direction

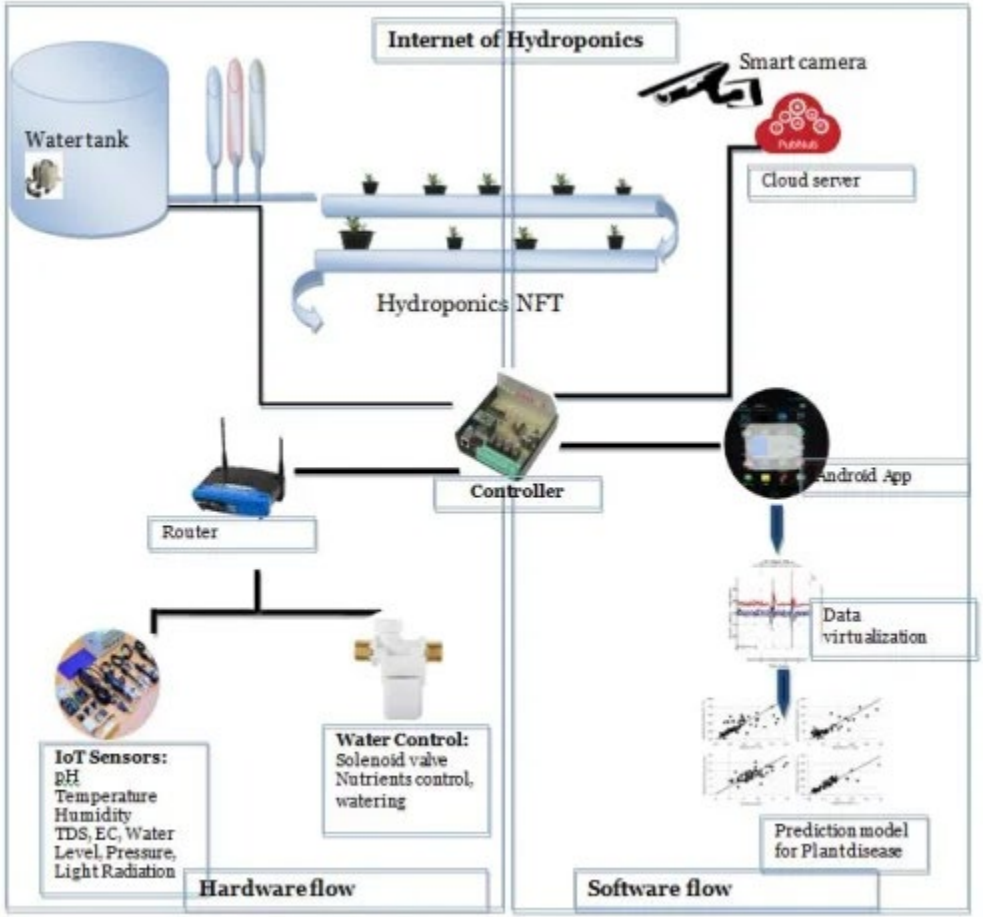
- High-Level Description of the overall system
- How will the system work
- What technologies are involved in building and operating the system
- Provide a baseline IoT and Software Architecture

f) Quality Management

- Define the characteristics of Quality that this project must meet.
- This definition of “Done” is used to check whether a developed item is qualified enough to meet the requirements of the client.
- ***Quality Assurance*** is the management of this process throughout the project.

g) Resources

- This section of the Proposal describes the materials, manpower and equipment gathered for the project.
- Can be simply divided into ***Human Resources*** and ***Non-Human Resources***.



Schematics of Internet of Hydroponics

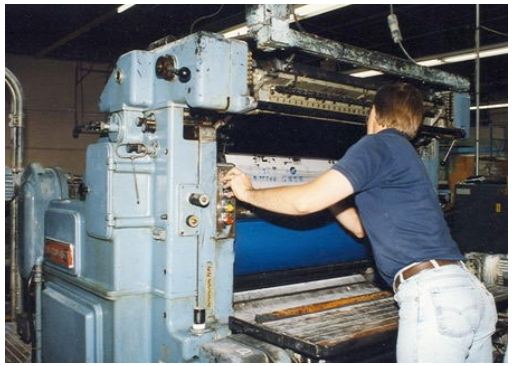
Case Study: IoT-Based Accident Detection & Response

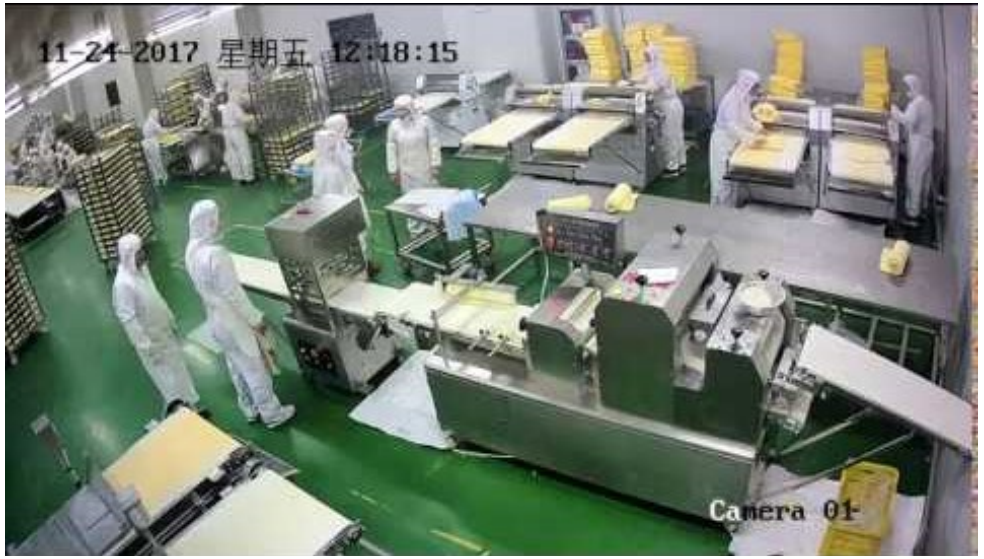
Situation:
Your team is tasked to meet an industry client who owns a factory. This facility has a long history of frequent accidents because it is a fast-paced working environment full of employees moving around heavy machinery.

All employees are only tracked when they punch into their shift. Afterwards, they proceed to all areas of the factory workspace, which spans a very wide area. An accident is only reported by employees who witness the occurrence and activate a general Alarm. Most injuries become fatal because emergency staff take too long to find out where the accident is, and which types injuries are expected.

They need a solution to this problem which does not interfere with the factory's physical workspace, not require additional infrastructure other than its present Wireless LAN and is informative to non-ICT personnel in the Occupational Safety and Hazard planning team.







Case Study Step 1: Problem Description

Identify the issues/problems:

- Employees are only tracked when the enter or leave the building (through punch card). They can only be identified when their Employee Card is read by the emergency staff.
- Accidents are only indicated by the general alarm triggered by Witnesses.
- The type of injuries can only be addressed when the Emergency Staff reach the location of the Accident and assess the victim/s.



Pain Points:

- Too much time passes between the occurrence of the Accident and Emergency Staff reaching the Victim, increasing the rate of fatality.
- Emergency Staff must guess where to go, based on the general direction of the Alarm.
- Emergency Staff must guess what injuries to prepare for, based on general direction of the Alarm.





What Issues should be clarified on?

Rough Solution:

An IoT system that monitor every employee and notifies Emergency Staff in case of a possible injury, based on who was affected, and in which part of the factory.

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Case Study Step 2: Scope

- We need a clear set of Scope Description (what does it do) and Objectives (what are the general, realistic goals).

- Scope must be a short description of the primary purpose/functions of the system to respond to the Problem Description. Example:
  - *This IoT system will involve the development of an activity tracker for each employee to track their general motion and orientation patterns.*

  - *In case an aggressive pattern is detected (indicating possible accident), it will notify the platform via WiFi and the Internet, along with the employee's identity and the identity of the connected WiFi Access Point (which will help identify the location of the accident).*

  - *The notification will inform Emergency Staff of who is involved, where the accident is located and what injuries to prepare for, based on the type of machinery at that location.*

  - *By collecting a history of accident occurrences, analytics can help identify hotspots to focus prevention measures.*

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Case Study Step 2: Scope (Cont.)

- The Objectives are attributes or general goals of the project, which addresses the Pain Points and Constraints.
- Must be S.M.A.R.T.
- Example:
  - *Near real-time notification of accidents upon detecting possible pattern indicating an accident.*
  - *Indication of accident location within 30 meters radius of an alarm station.*
  - *Advice on injuries expectations consider the machines at accident location.*
  - *Implementation consume none of the ground workspace and minimize obstruction of employee work.*
  - *Rely on existing network to facilitate communications.*

*From here onwards, determining **Step 3: Stakeholders** then setting the **Step 4:** Schedule (**Timeline**) and **Deliverables** should be straight forward.*

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Case Study Step 5: Solution Direction?

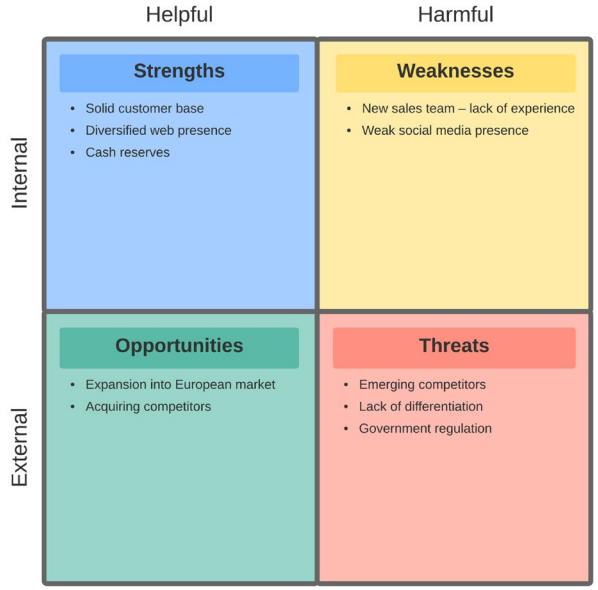


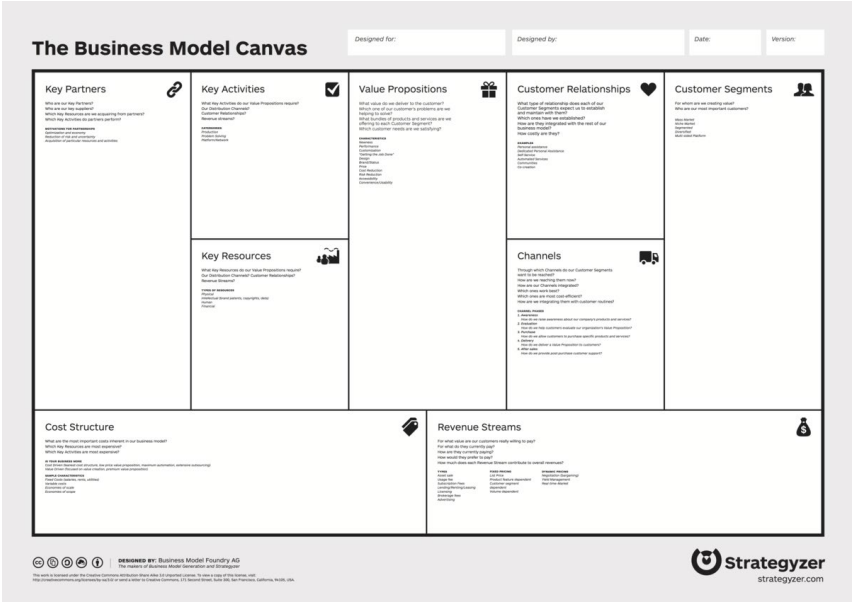
- Derive Potential Solutions. Here are some pointers:
  - Identifying the Problem Domain – tap into knowledge of this domain
  - Choose technologies commonly used in this domain
    - Avoid technology gaps
    - Make use of existing and tested tools/techniques for these kinds of projects.
  - Use the ***Project Forces*** to describe the Solution Direction
  - Visualize your Potential Solutions using High-Level Designs (block diagrams and flowcharts are useful).
  - Try to think outside the box, but not outside the constraints!
  - Evaluate the **Feasibility** of each Potential Solution, according to the Constraints

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Case Study Step 5: Solution Direction (C

- *Now, choose a Solution:*
  - *Apply the Objectives as a criteria to select one from the pool of Potential Solutions.*
  - *Perform Risk Analysis – to identify the threats to the project*
    - *One example way is KoST – identify Knowledge, Skills, and Technology Gaps*
  - *Cost-Benefit Analysis – which is the best for the client?*
  - *Feasibility Analysis – is this solution possible and realistic?*
  - *Some projects involve defining a **Business Case** for the chosen solution (e.g., Business Model Canvas)*
- If you cannot decide, that means you have not gotten enough information yet!



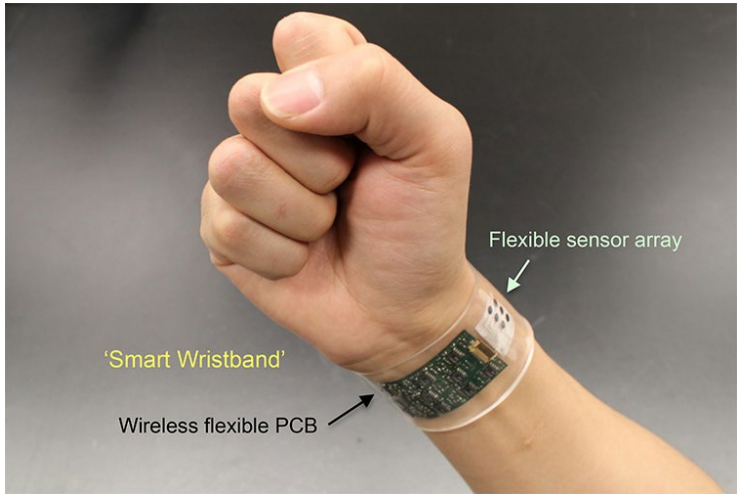


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Case Study Step 5: Solution Direction (C

- Now back to the Case Study! Example:
  - Problem Domain: Activity Tracking
  - Potential Solutions:
    - Vision-Based Activity Tracking using Kinect – Rejected! Involves heavy installation in the workspace.
    - Mobile App – Rejected! From Client interview, it appears that Employees are not allowed to bring personal devices into the facility.
    - IoT Machine Monitoring System – Plausible! Installation of vibration sensors and status monitoring of machines into remotely connected nodes. Uses AI to learn the common operation patterns of machines so anomalies indicate accidents. But Rejected! due to wide Technology Gap (need to incorporate advanced Machine Learning) and short development Time.
    - Smart Wearable Activity Tracker – Accepted! The implementation is non-obstructive as it is worn as a watch by employees.





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Risk Drivers using KoST Gap Analysis

- In any project, there is the potential for difficulty caused by lack of knowledge, skills and appropriate tools
- It is important at the outset of a project to consider this
- One approach is called “KoST” analysis
- Note : This is not “famous”, and you will not easily find references to it! But it is a useful first look at project risks – one which, if not carried out, could lead to disaster!
- (In the Myki project in Victoria, of the 6 tendering providers, 5 had the experience of a successful smart-ticketing system implementation; the contract was awarded to the sixth provider! Enough said!!)

Principal risk drivers (KoST):

- Knowledge Gap (*don't know*)
- Skill Gap (inexperience)
- Technology Gap (unknown/young or unavailable)

KoST analysis attempts to determine

- knowledge and skill gaps a team/individual has, and
- whether any technology gaps exist.

Note that in general there will be many other risks associated with a project, and KoST is not a full risk analysis – covered in a later lecture.

- Knowledge Gap
  - Lack of knowledge of the problem domain, and the particular context of the desired solution
- Skill Gap
  - Lack of the skills needed to implement the system according to the proposed approach
- Technology Gap
  - Using a technology that is new, without certainty that it will work for the proposed solution {more than just a skill gap with the technology, but also an uncertainty about whether the technology can actually work in this context}

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Architecture Design

- **Systems Modelling** is a process of conceptually visualizing the ***structure*** and ***behavior*** of a system.
- There are various methodologies to do System Modelling. In Software, the most prominent way is the **Object-Oriented Approach**.
- The output of this process is a set of documentation that describe the proposed system's ***structure*** and ***behavior***, called Architecture Design.
- There are 2-Levels of Architecture Design:
  - High-Level Design
    - Shows the Overall Structure but not the exact mechanism in each module.
    - Translated from the main Use-Cases / Usage Scenarios.
    - Also based on meeting non-functional requirements (e.g., reliability, throughput, consistency, etc.).
    - Normally do not change.
  - Low-Level Design
    - Shows the detailed mechanism and components structure in each module.
    - Must account for Programming Language, Platform, Technology and Hardware.
    - Must be detailed enough to be implemented using Programming Language, Platform, Technology and Hardware.

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High-Level Architecture

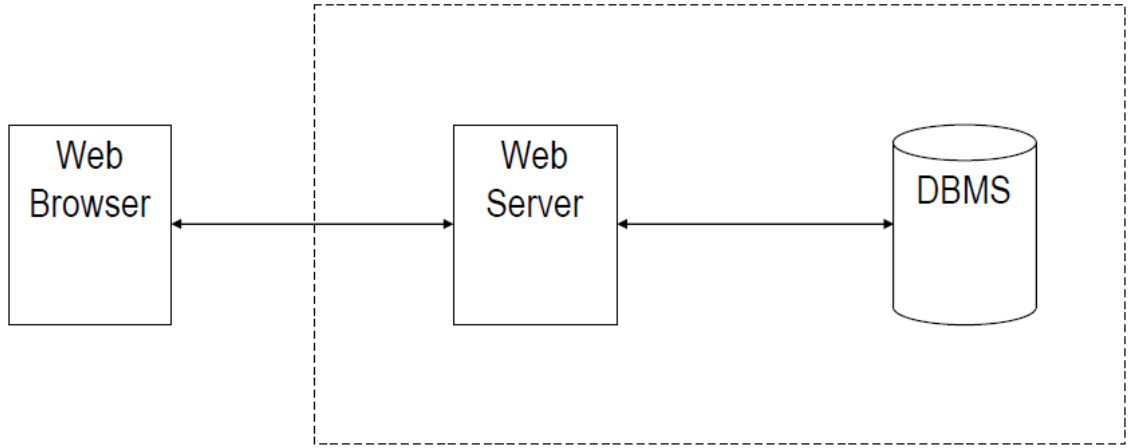
- This must be present in the **Project Proposal** to inform the Client that this is what the proposed solution will look like.

- It Captures the gross structure of the system:
  - How it **consists** of software and hardware components
  - How the **interactions** between components take place
  - What are the key **properties** of the components

- Why is this important?
  - Handling **Complexity**
  - **Communication** among Stakeholders
  - Early **Design Decisions**
  - Architecture is **Transferable** and **Reusable**

- A good Architecture Design:
  - Uses consistent set of principles and techniques
  - Resilient to changes
  - Source of guidance across the product timeline
  - Reusable

■ Example: Web application



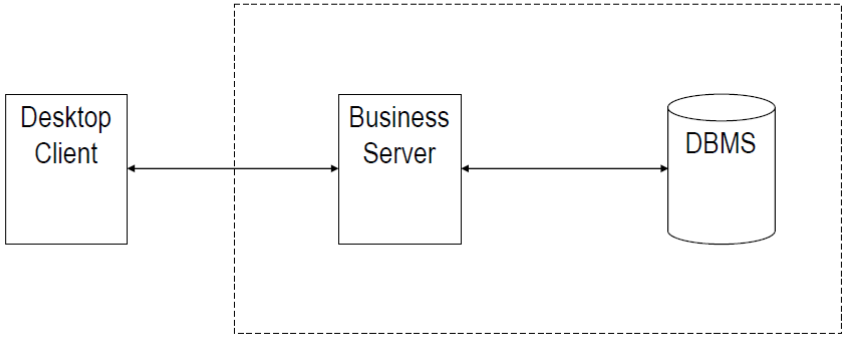
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High-Level Architecture (Simple Software Examples)

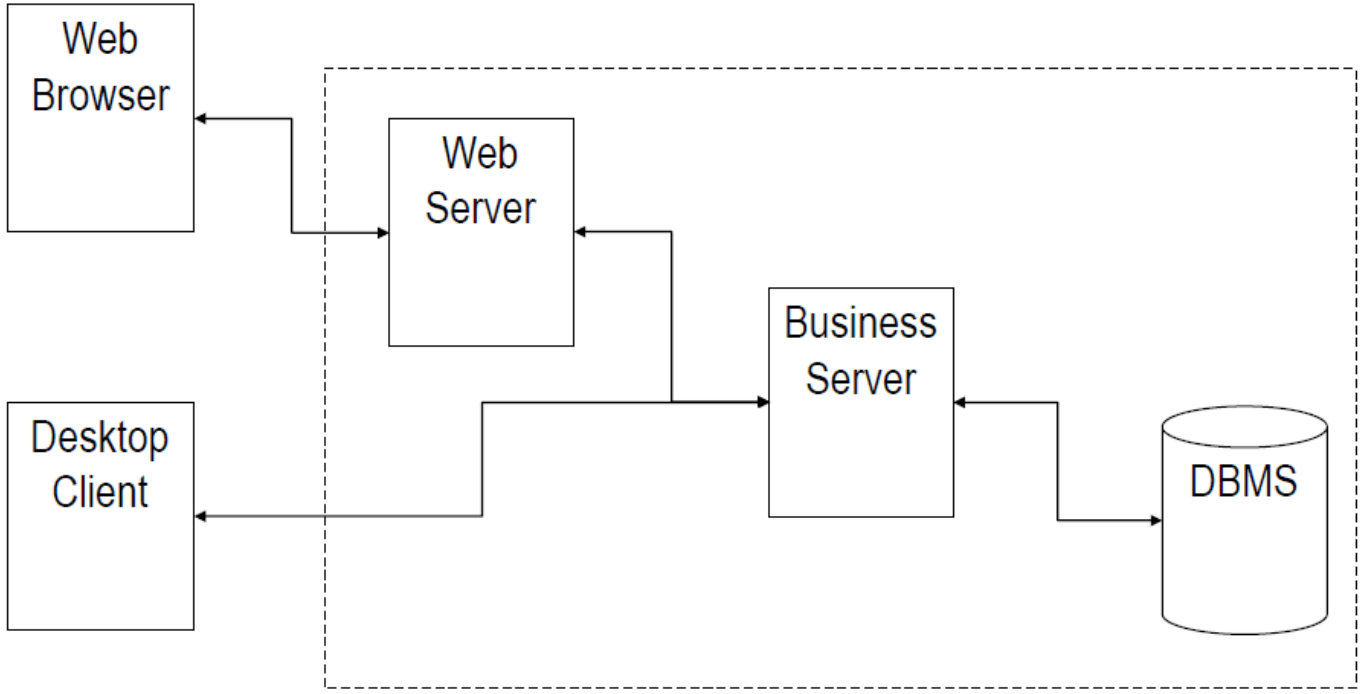
- Example: Desktop application (old days)



- Example: Desktop application

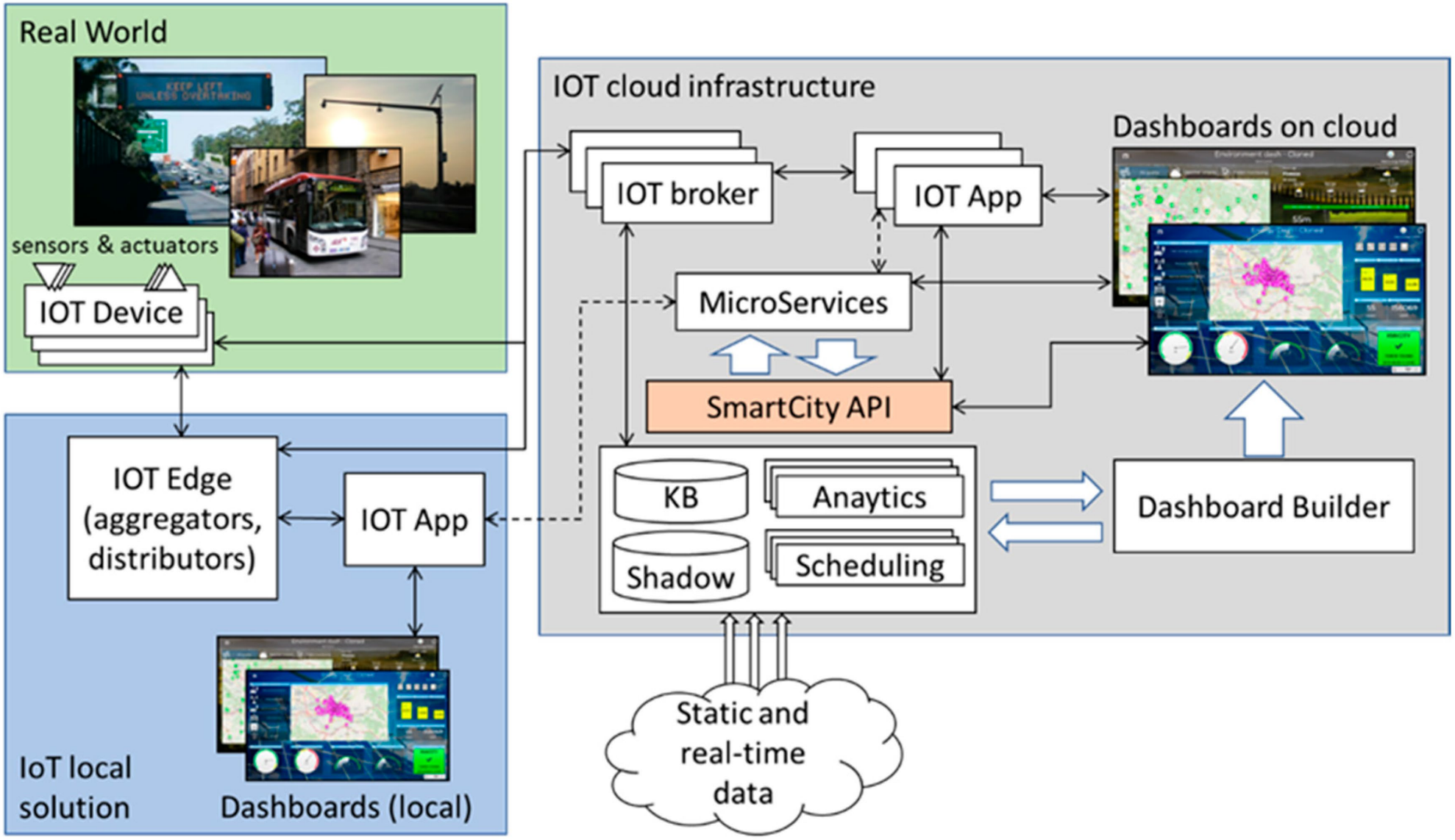


Enterprise Framework

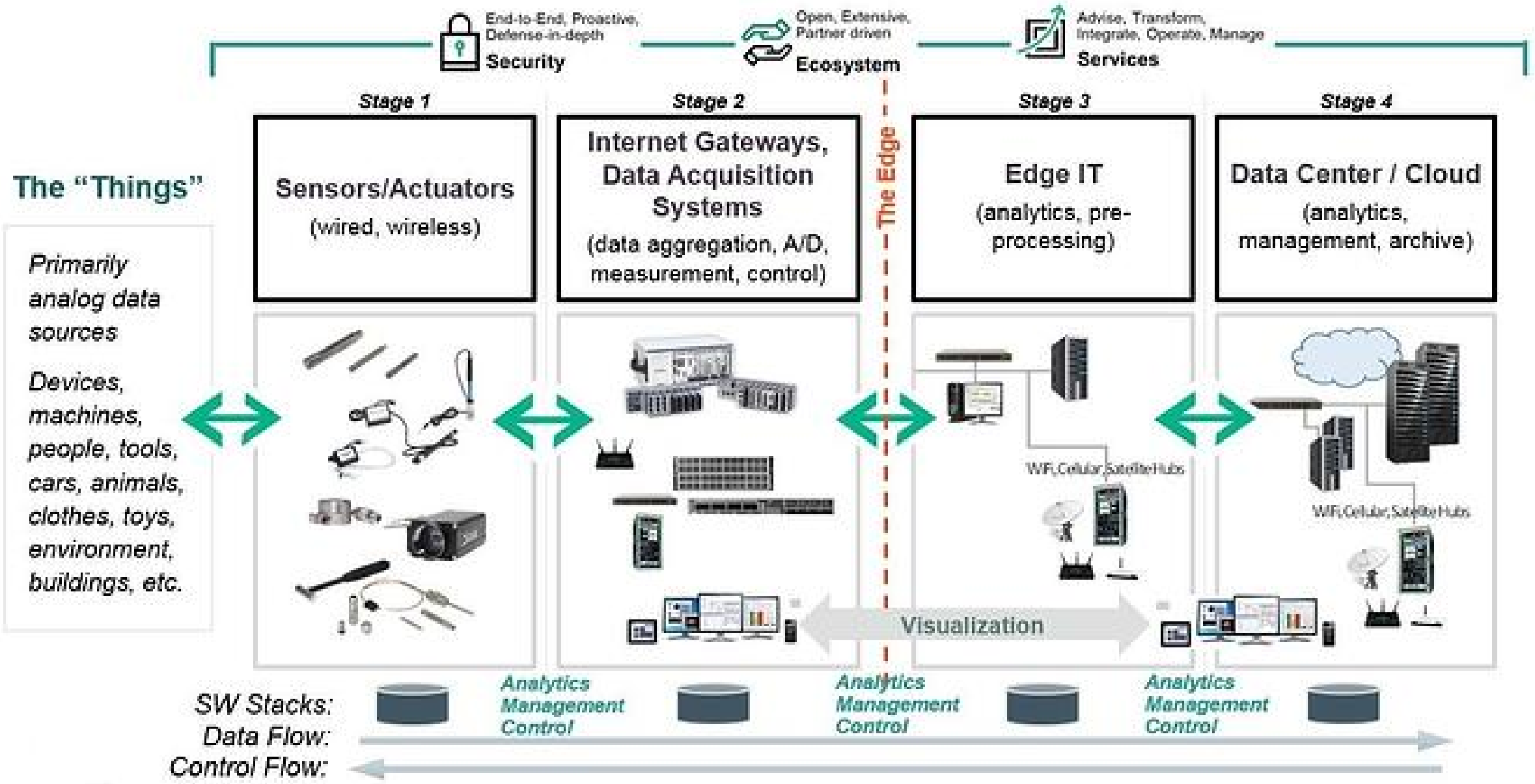


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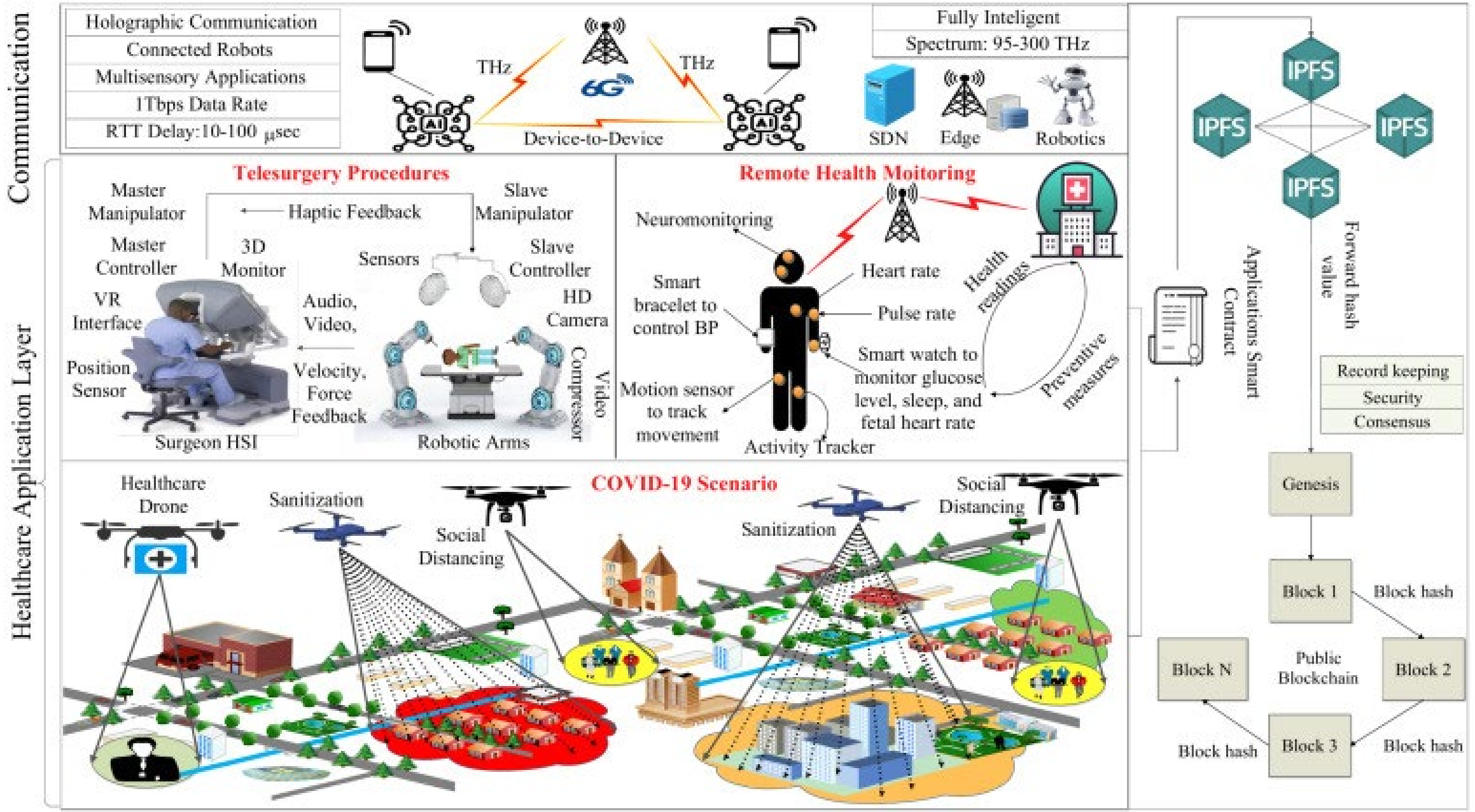
High-Level Architecture (IoT Examples)



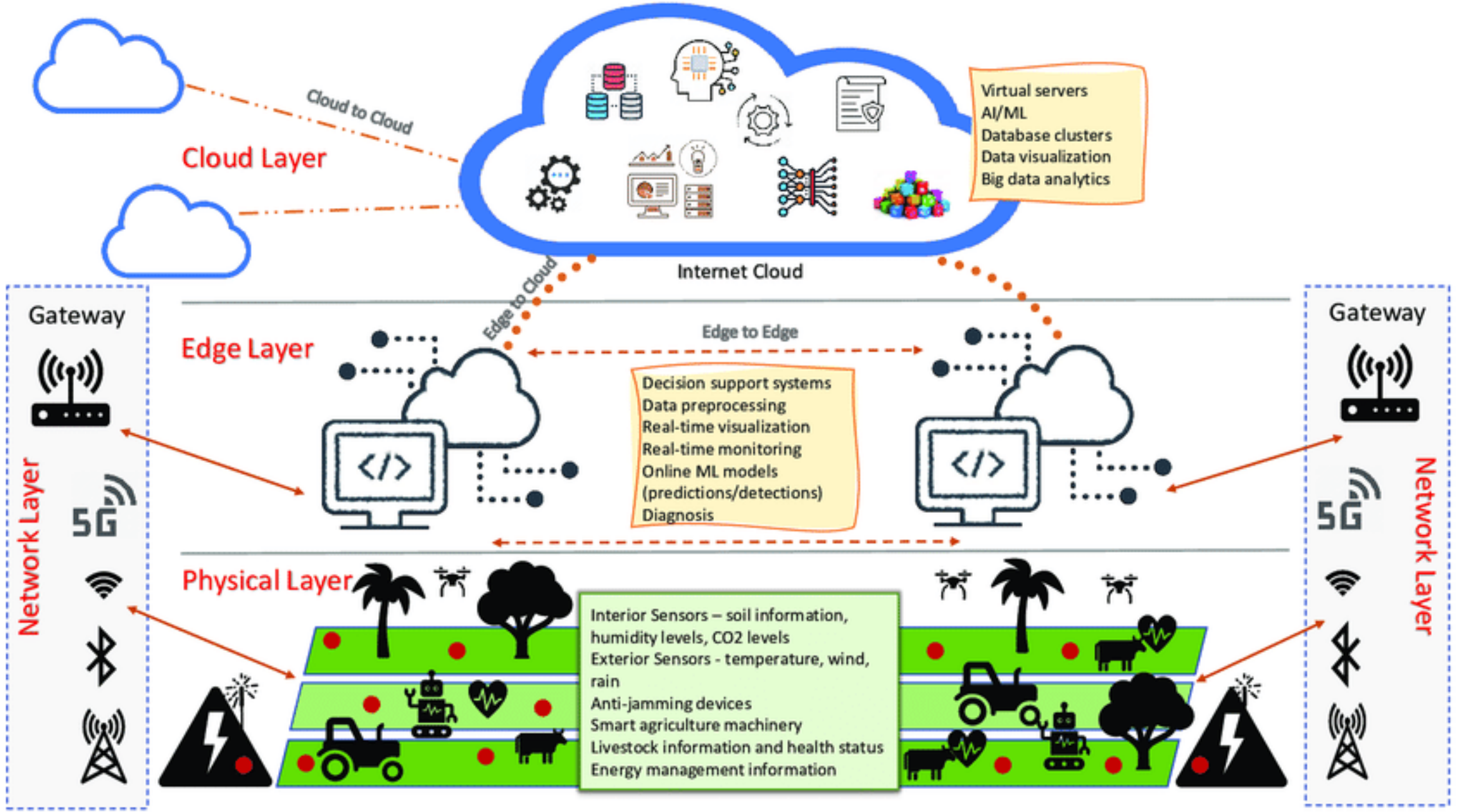
High-Level Architecture (IoT Examples)



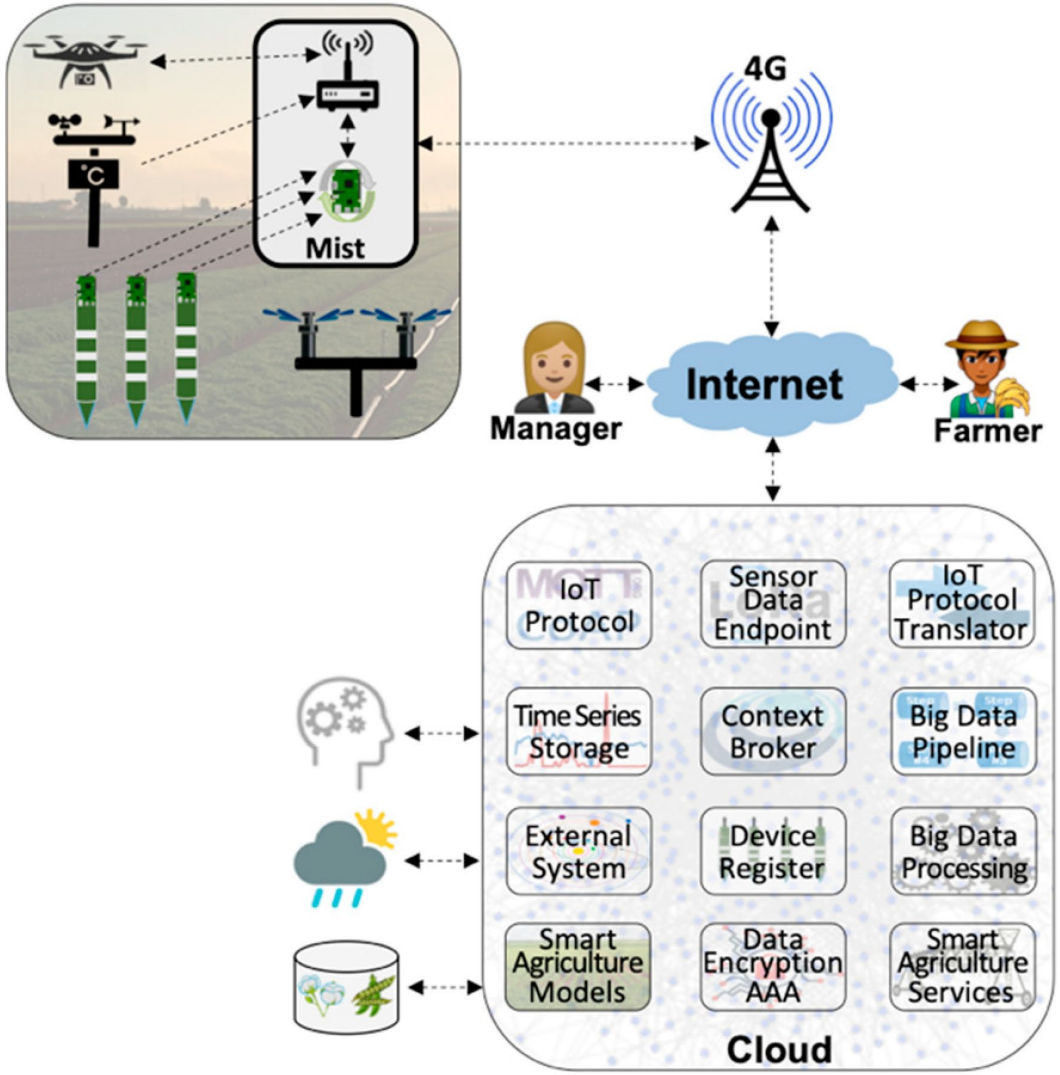
High-Level Architecture (IoT Examples)

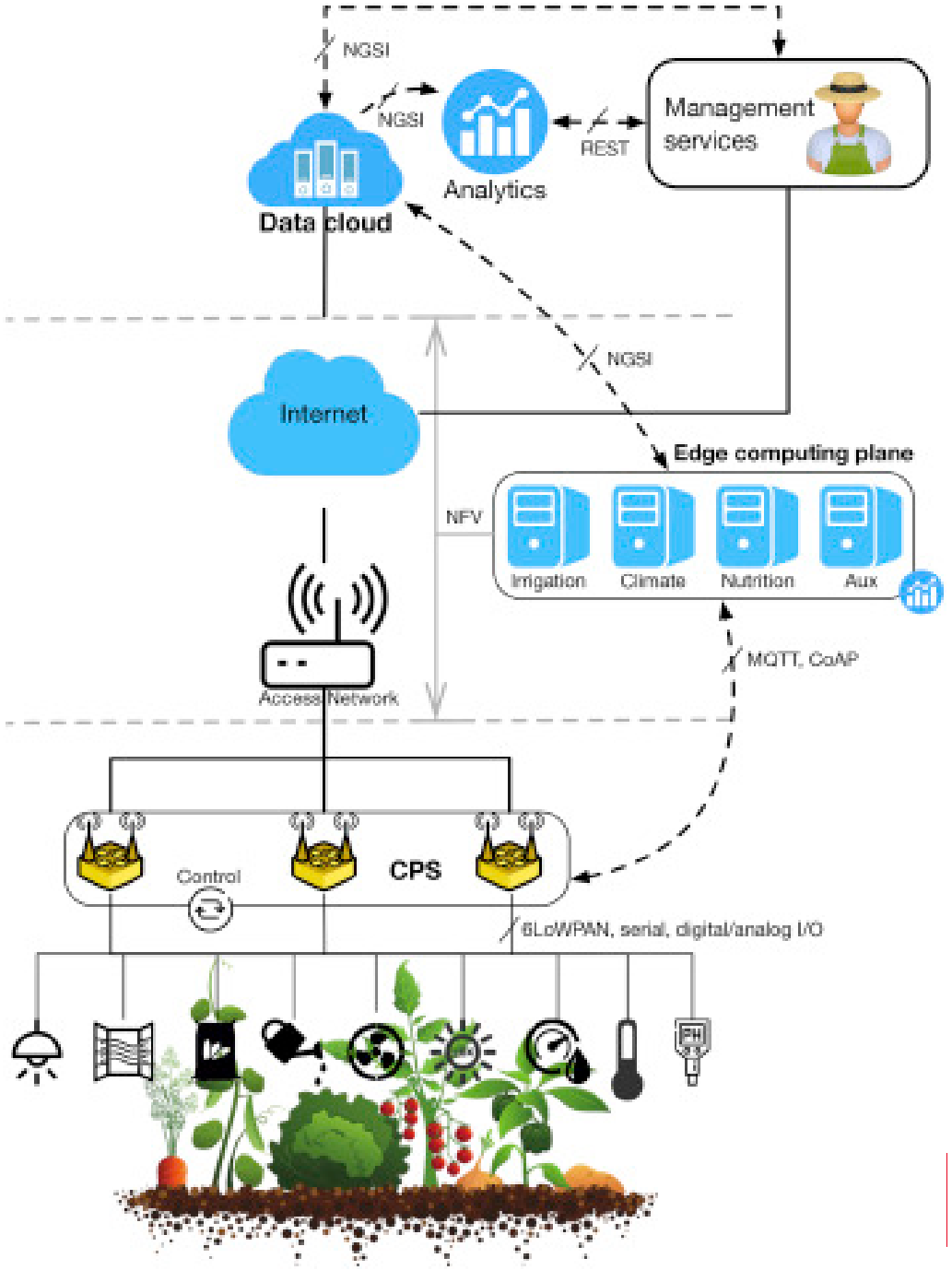


High-Level Architecture (IoT Examples)

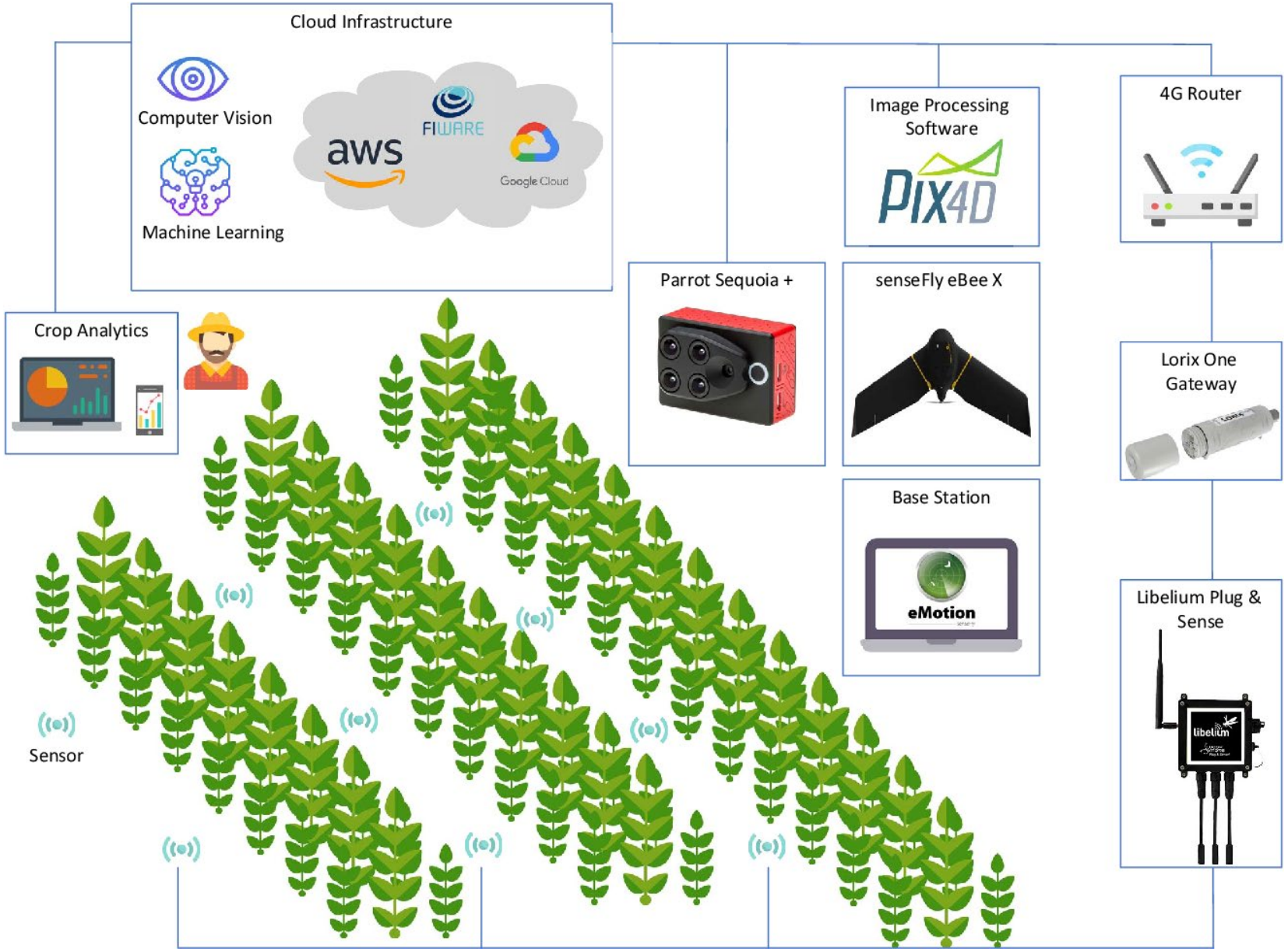


High-Level Architecture (IoT Examples)





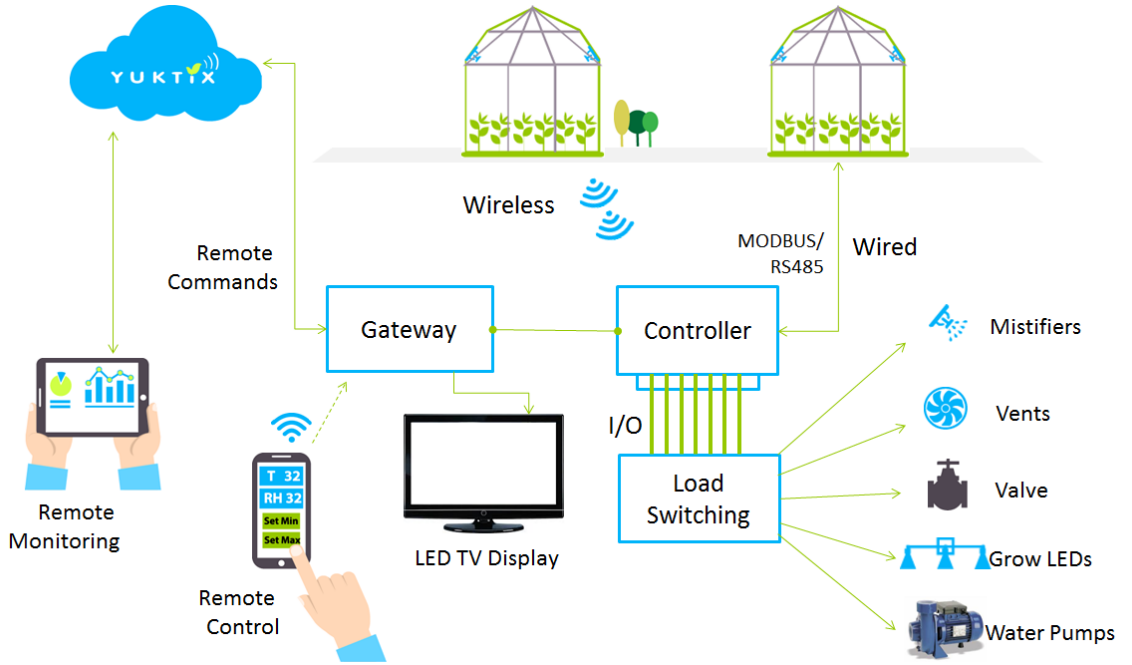
High-Level Architecture (IoT Examples)

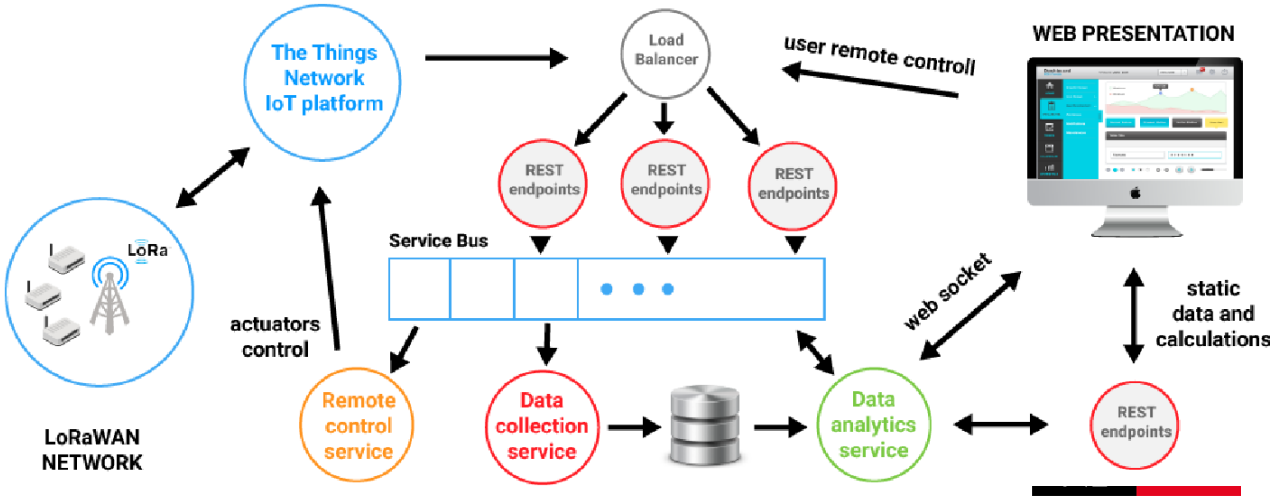


High-Level Architecture for IoT

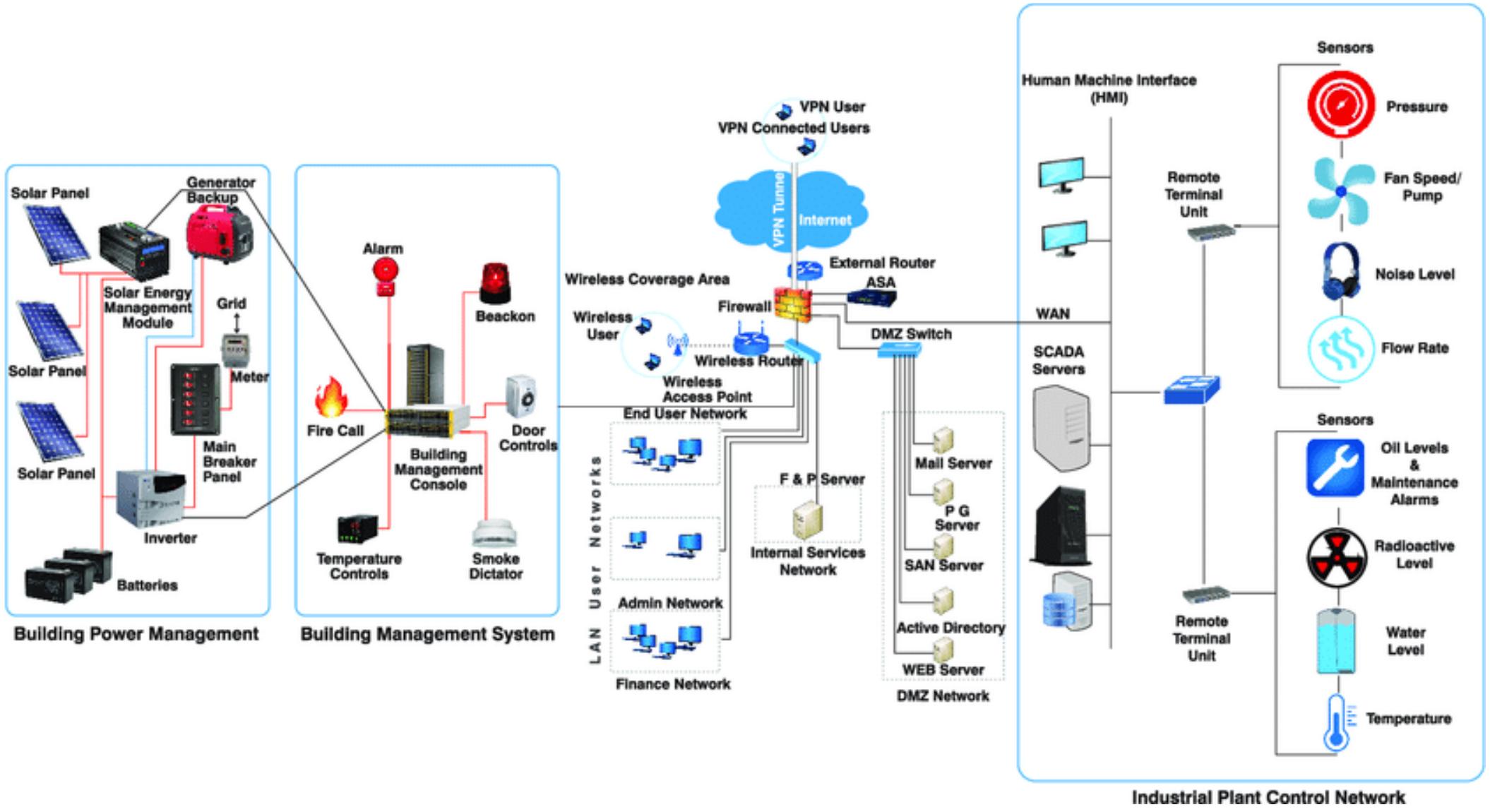
- Notice that there are no set standard for describing the High-Level Architecture Design for IoT system, because IoT systems is a Multidisciplinary (involves a combination of Computing, Engineering and Construction disciplines).

- Common practices:
  - Use a separate Block Diagram to represent the Hardware, Software and Network Architectures:
    - **Hardware**: Sensors, Actuators, Modules, MCU Platforms, Interfaces, Edge Devices, User Devices, etc.
    - **Software**: Communications Protocols (like MQTT, CoAP), Database, Cloud Platform, Services, etc.
    - **Network**: Networking Devices, Subnets, Gateways, End Devices, ISP, Cloud, etc.
  - Elaborate using the **"4+1" Views of Architecture**.

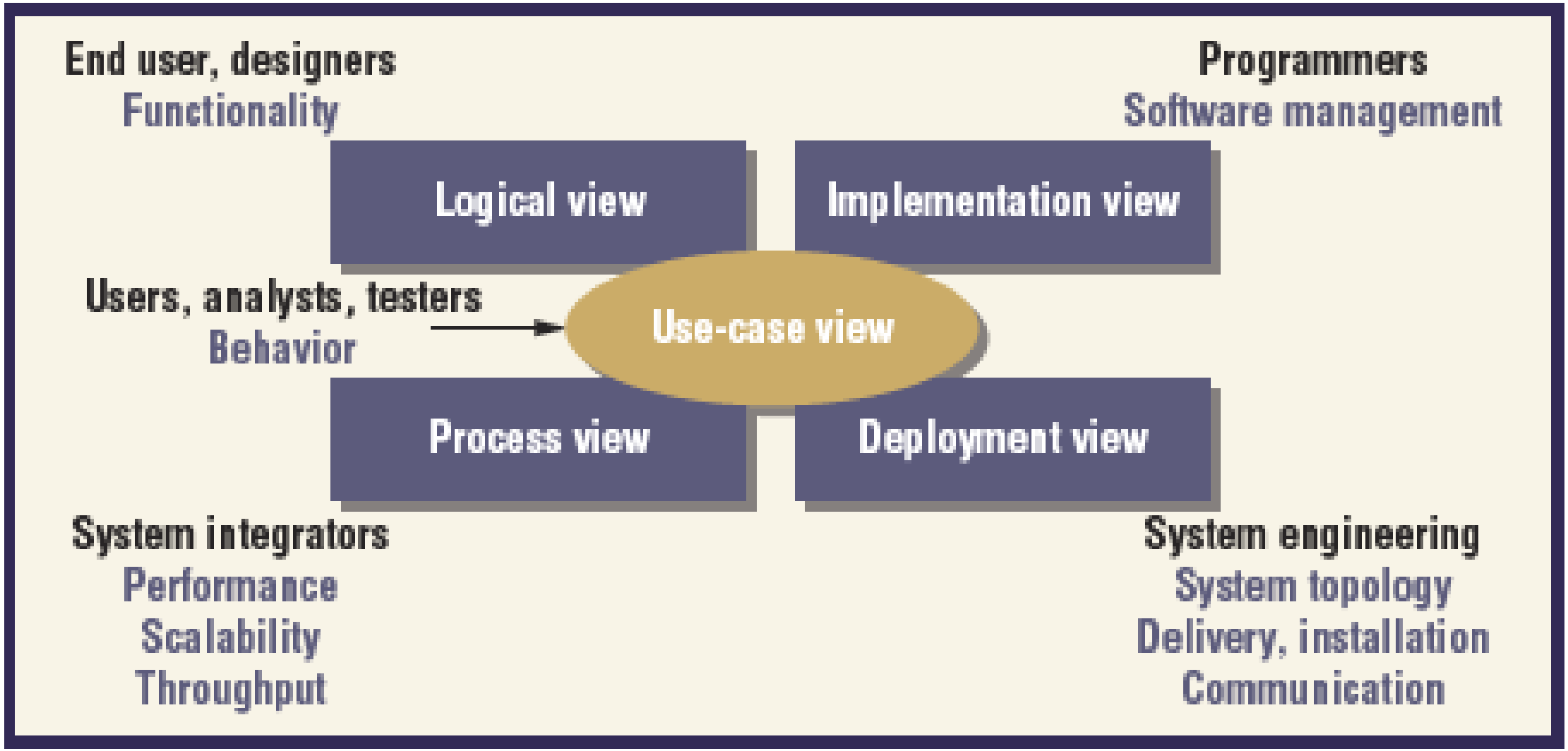




High-Level Architecture for IoT (Cont.)



The 4+1 Views of Architecture



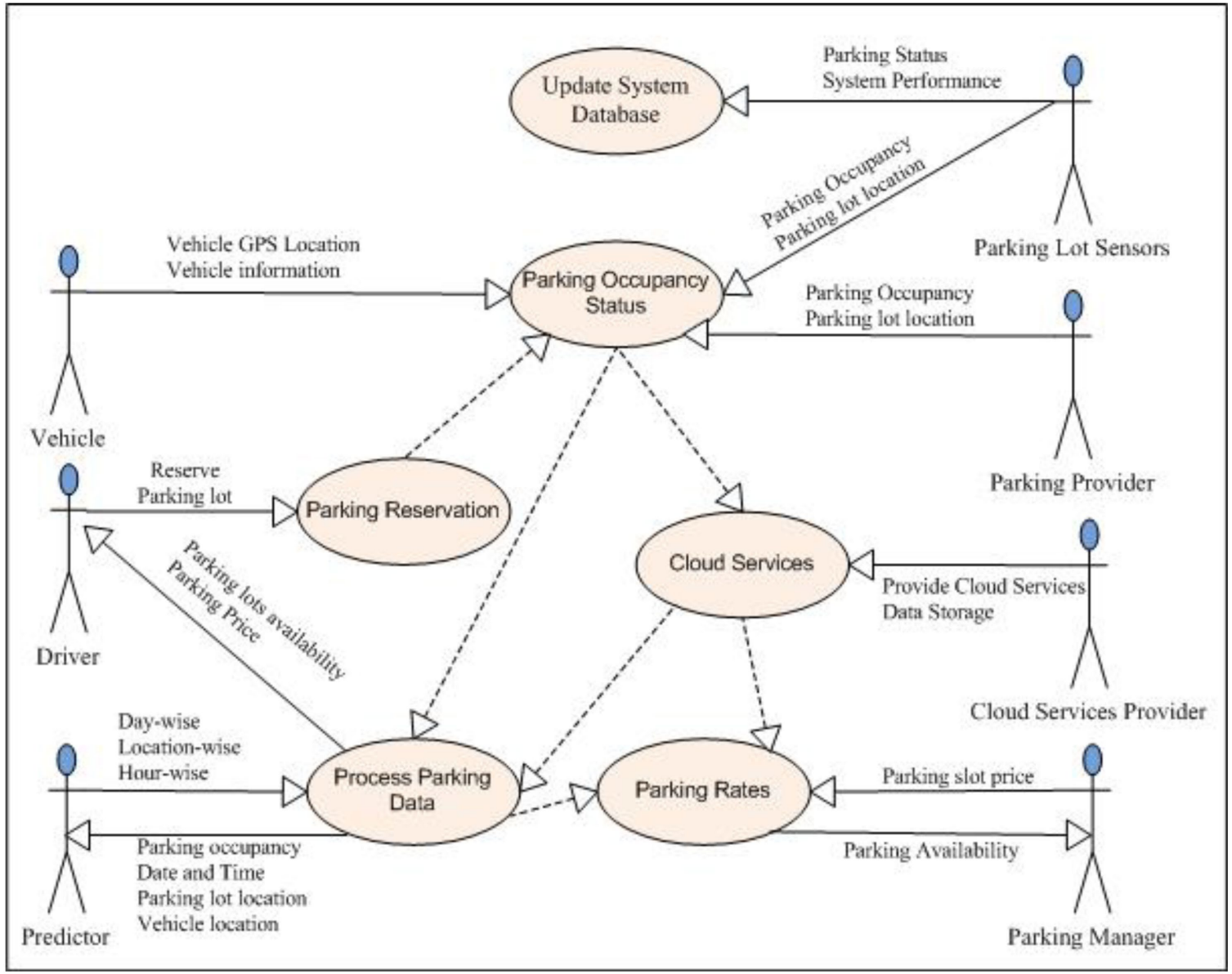
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Use Case View

- The Use Case View shows how the Stakeholders interact with the System – translated from the Use Cases (elaborated through the Usage Scenarios).

- It also the relationship between Functions/Operations.

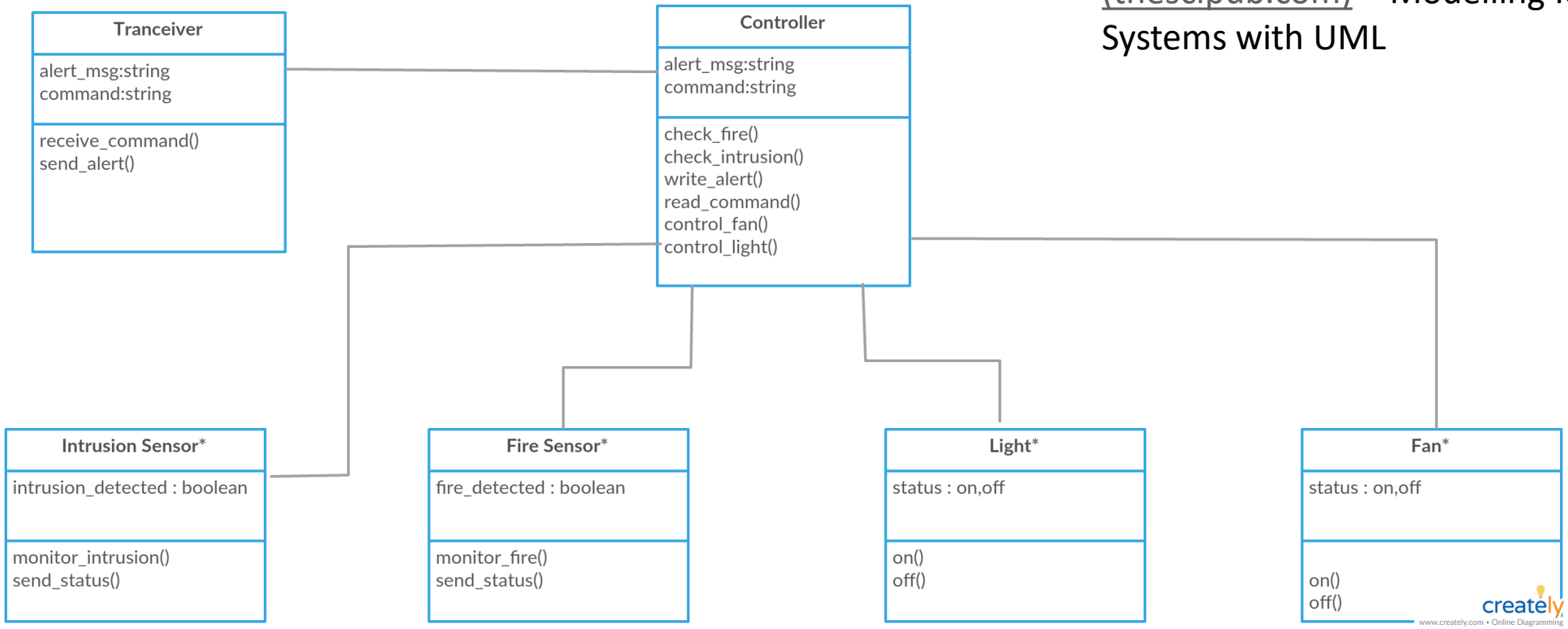
- Learn more here:
  - UML Use Case Diagram Tutorial | Lucidchart
  - What is Use Case Diagram? (visual-paradigm.com)



Logical View

- The Logical View is a subset of the Design Model which presents architecturally significant design elements
- Describes the most important classes
- Their organization in packages and subsystems into layers
- It also describes the most important use-case realizations, for example, the dynamic aspects of the architecture

- Class Diagrams are one of the most useful ways to express the Logical. Design of a system.
- Learn more about them here:
  - UML - Class Diagram (tutorialspoint.com)
  - ajeassp.2021.81.93.pdf (thescipub.com) – Modelling IoT Systems with UML

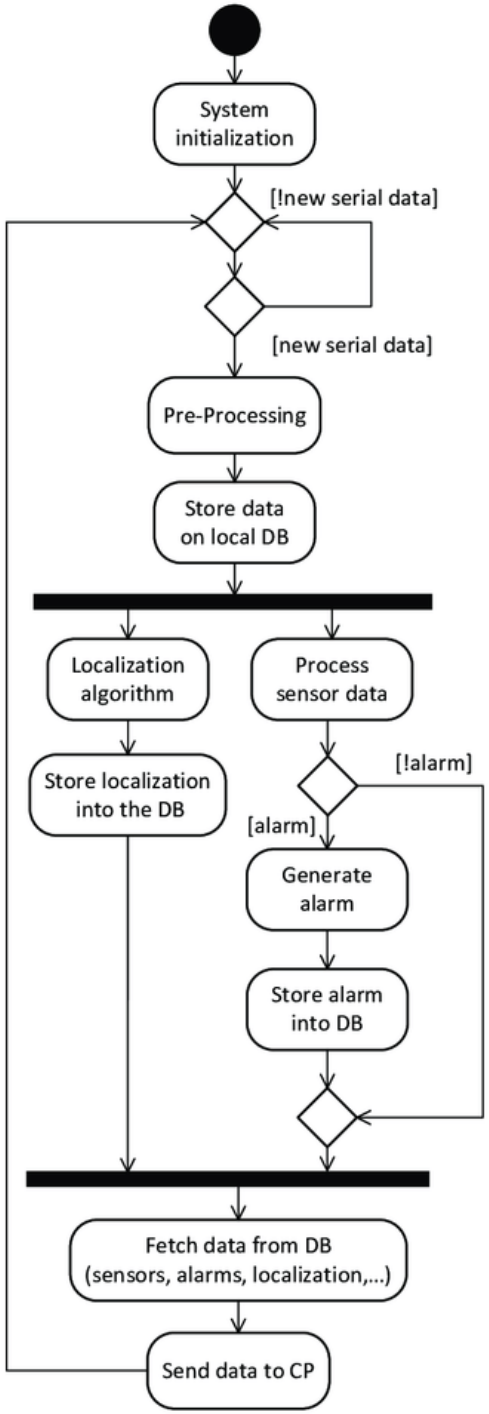


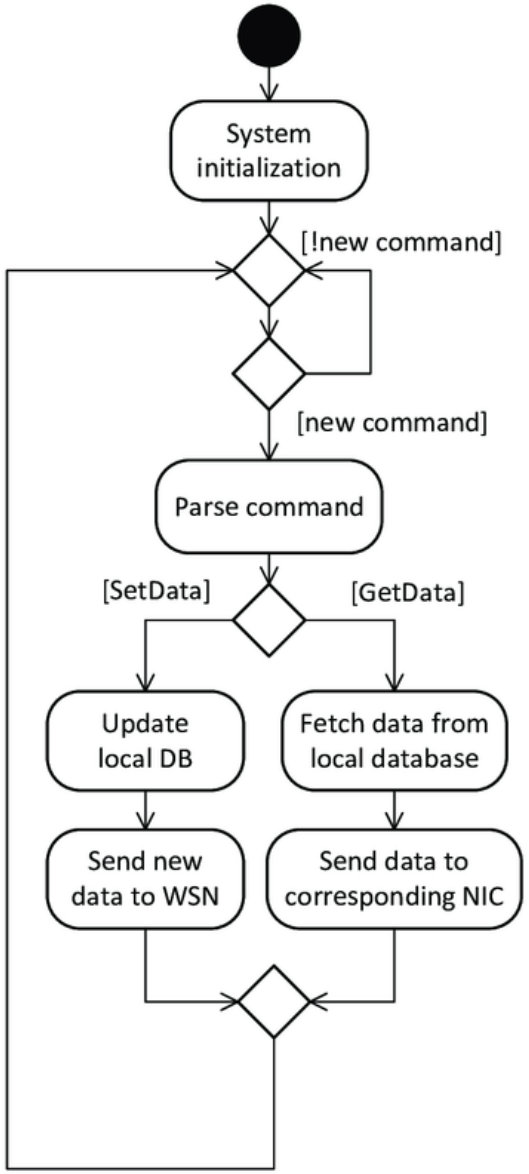
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Process View

- Consists of the ***processes*** and ***threads*** that form the system's ***concurrency*** and ***synchronization*** mechanisms, as well as their ***interactions***
- Is captured using class, interaction and state transition diagrams with a focus on active classes and objects.

- Learn more about Activity Diagrams here:
  - UML - Activity Diagrams (tutorialspoint.com)
  - What is Activity Diagram? (visual-paradigm.com)



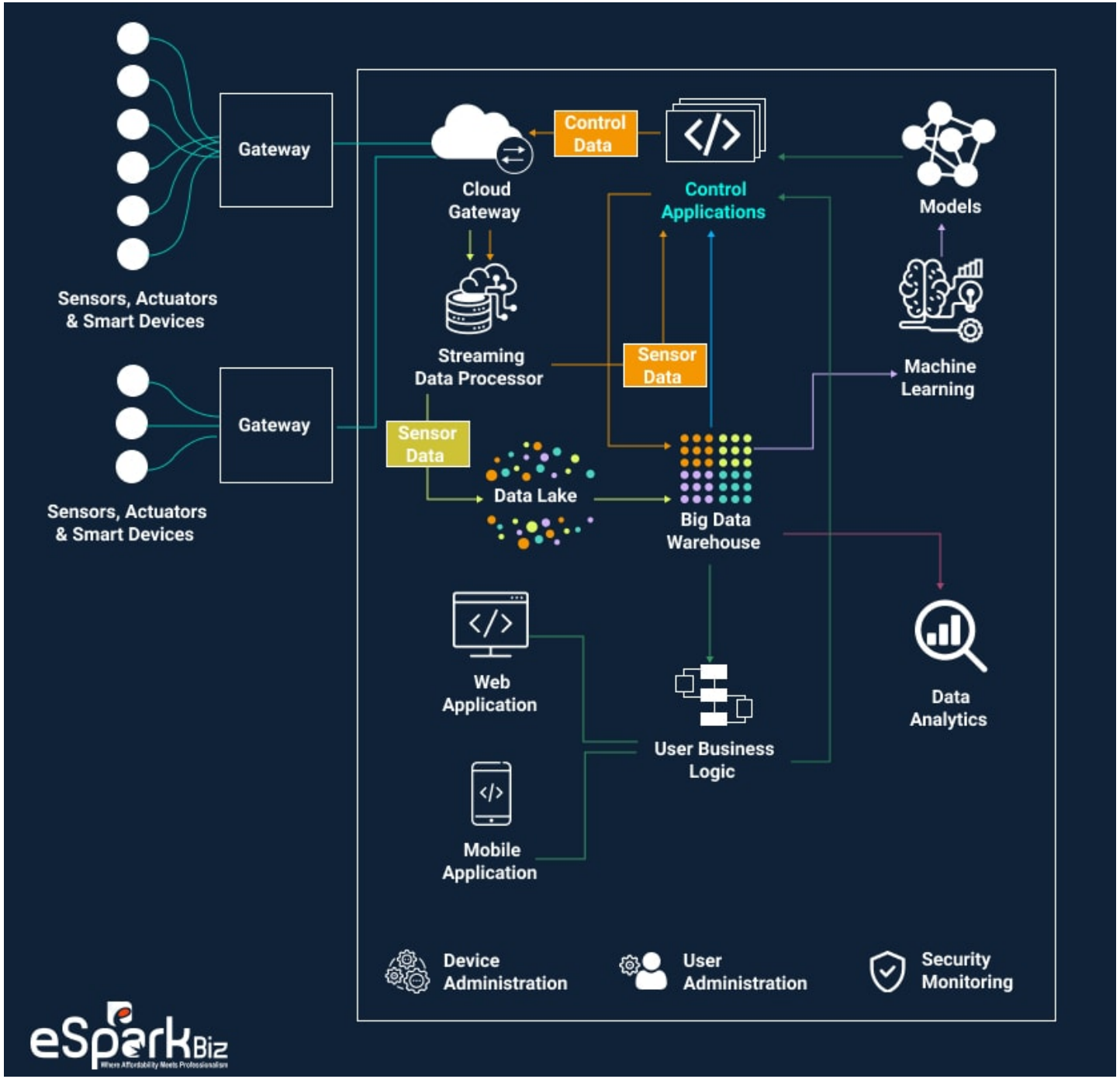


Deployment View

- The Deployment View in Software Development is meant to describe the hardware components and networking involved in the proposed system.

- In IoT Projects, this view is represented by the 3 Block Diagrams:
  - Hardware
  - Software
  - Networking

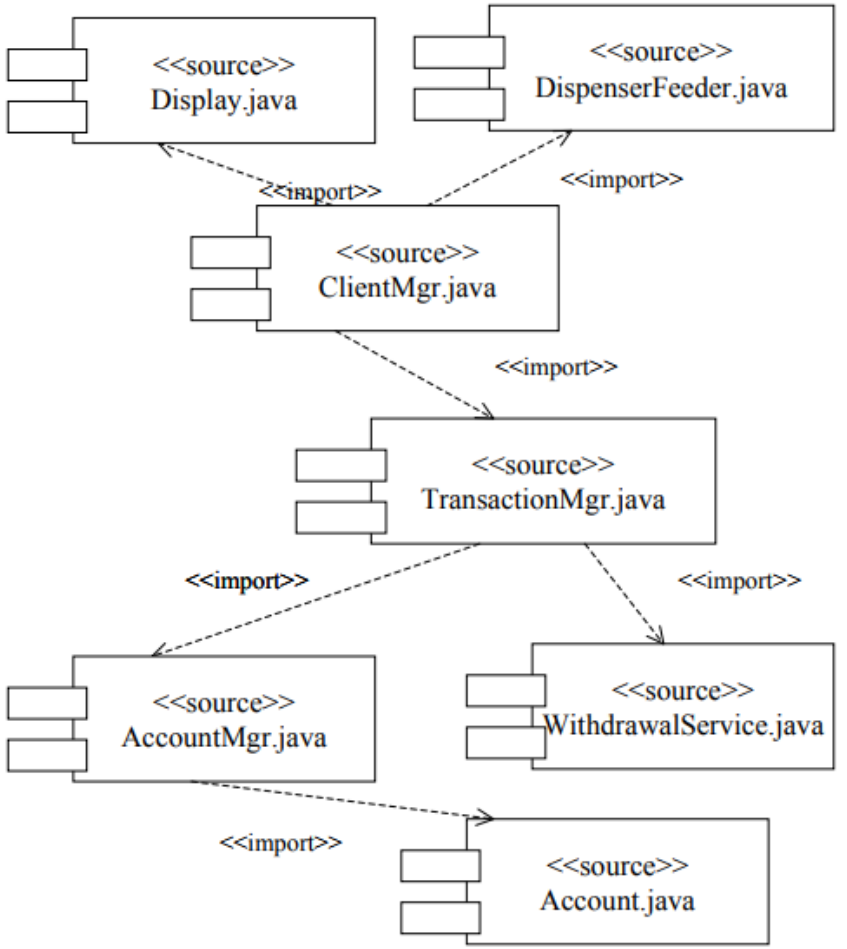
- Moving On!



Implementation View

- Describes the **organization of static software modules** (source code, data files, executables, documentation etc.) in the development environment in terms of **Packaging and layering**
- Are modelled using UML Package Diagrams.
- UML components are physical and replaceable parts of a system that conform to and provide the realization of a set of interfaces

Classes	Source Components
CardReader, Display, KeyPad	Display.java
DispenserFeeder, DispenserSensor, CashCounter	DispenserFeeder.java
ClientMgr	ClientMgr.java
TransactionMgr	TransactionMgr.java
AccountMgr	AccountMgr.java
WithdrawalServic e	WithdrawalService.java
Account, Persistent class	Account.java



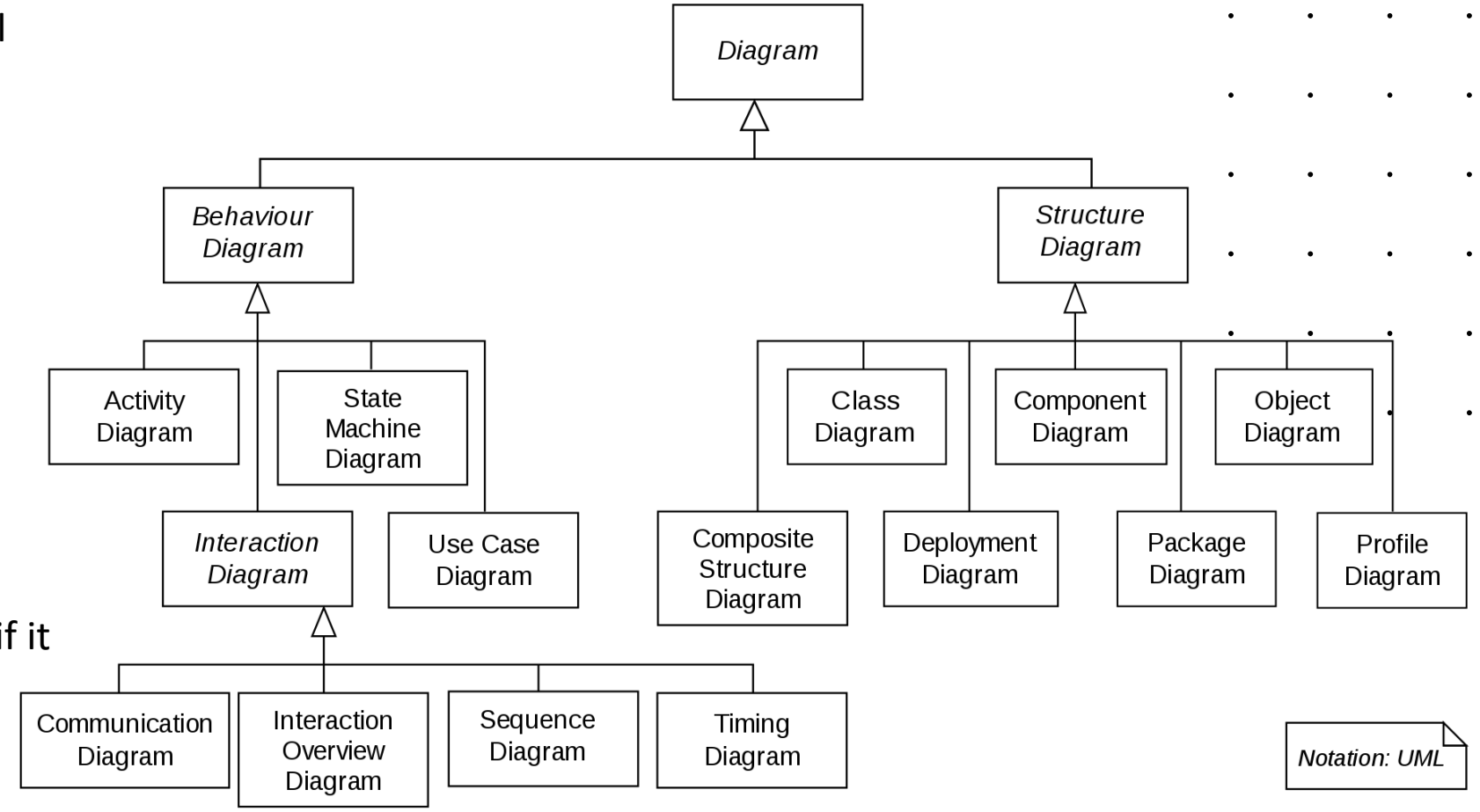
- However, in IoT Projects, the Implementation View is often represented by the Software Block Diagram.
- Moving On!

UML

- The Unified Modelling Language is a standard set of diagrams that all software engineers can use for building and understanding Architecture Designs.

- Learn about them here:
  - What is Unified Modeling Language (UML)? (visual-paradigm.com)
  - UML Tutorial (tutorialspoint.com)

- You can use diagramming tools if it supports UML 2.x, such as:
  - Visual Paradigm
  - Draw.IO
  - StarUML
  - Microsoft Visio
  - LucidChart



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Takeaway for this Week

1. Follow the steps in IoT Project Proposal process to identify your “Scope” of IoT work.

2. Your IoT System Design consists of 2 stages of Diagrams:
   A. High-Level Architecture
      - Usually in 3 diagrams: Hardware, Software and Network Architecture

   B. Low-Level Architecture
      - Circuit Diagram
      - Flowcharts
      - Use Case, Class, ERD, or etc. diagrams needed for software design

3. Pick one of the tools available and look for online tutorials on how to use them to draw all the diagrams you need.

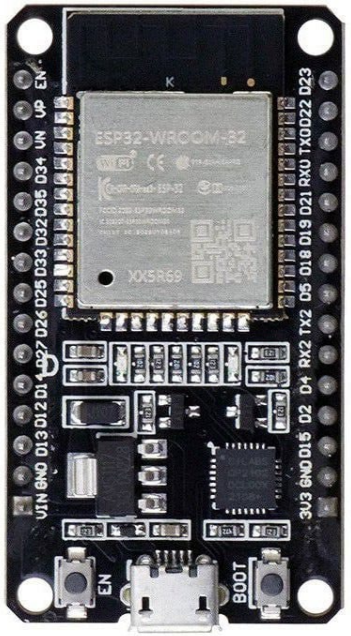
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IoT Track – Week 2

ESP32 – The IoT Microcontroller Development Platform

Prepared by Dr. Mark Tee Kit Tsun (SUTS)

Introduction to NodeMCU ESP32



- ESP32 is the name of a microcontroller that has the ability to connect to WIFI (and Bluetooth).

- NodeMCU ESP32 is an electronics development board (like Arduino Uno and M5Stack).

- Important specifications to note about this board:
  - It has WIFI and Bluetooth capability (Important for small IoT applications)
  - Operates on 3.3V (but you can safely power it via USB)
  - Has many pins with various functions (but not all pins can be used)
    - 18 Analog-to-Digital Converter (ADC) channels
    - 3 SPI interfaces
    - 3 UART interfaces
    - 2 I2C interfaces
    - 16 PWM output channels
    - 2 Digital-to-Analog Converters (DAC)
    - 2 I2S interfaces
    - 10 Capacitive sensing GPIOs

Important Task 1:
Study these tutorial pages to learn how to setup with Arduino IDE:
https://randomnerdtutorials.com/getting-started-with-esp32/

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Digital Input & Output using NodeMCU ESP32

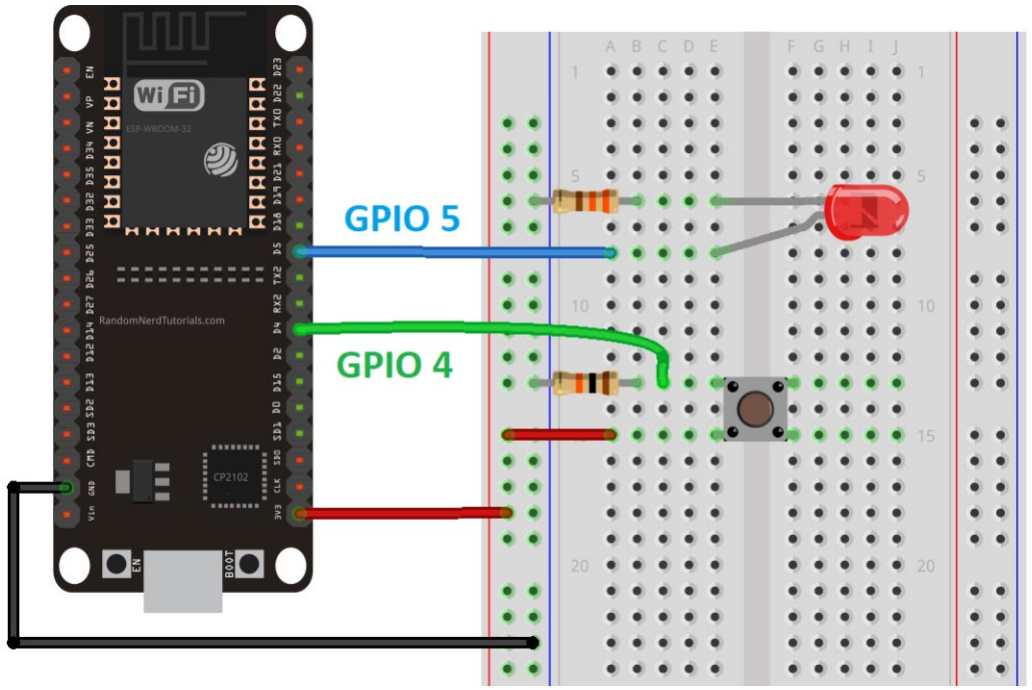
Mastery of the ESP32 begins with knowing how to do:

A. Digital Input (Sensors)

- Reading the pin to see if the connected component is sending a HIGH/LOW signal
- Example applications: Buttons, Switches, IR Sensors, Etc.

B. Digital Output (Actuators)

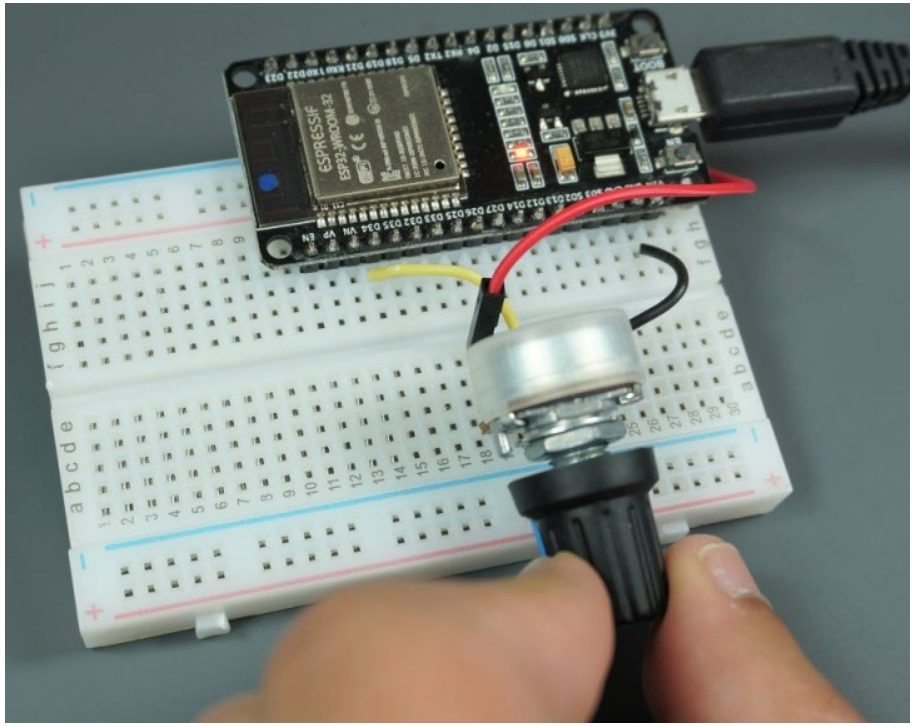
- The ability to output 3.3V (HIGH) or 0V (LOW) voltage from the pin
- Example applications: LED lights, Relay Switches, Etc.



Important Task 2:
Use this Tutorial and the components in your Kit to learn how to utilize LEDs and Buttons with your ESP32:
https://randomnerdtutorials.com/esp32-digital-inputs-outputs-arduino/

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Analog Input using NodeMCU ESP32



- Microcontrollers are essentially digital computers, so they only understand the two binary states: ON or OFF.
  - Equivalent to HIGH/LOW or 3.3V/0V.
  - It does not know how to measure 2.5V or 1.2V.

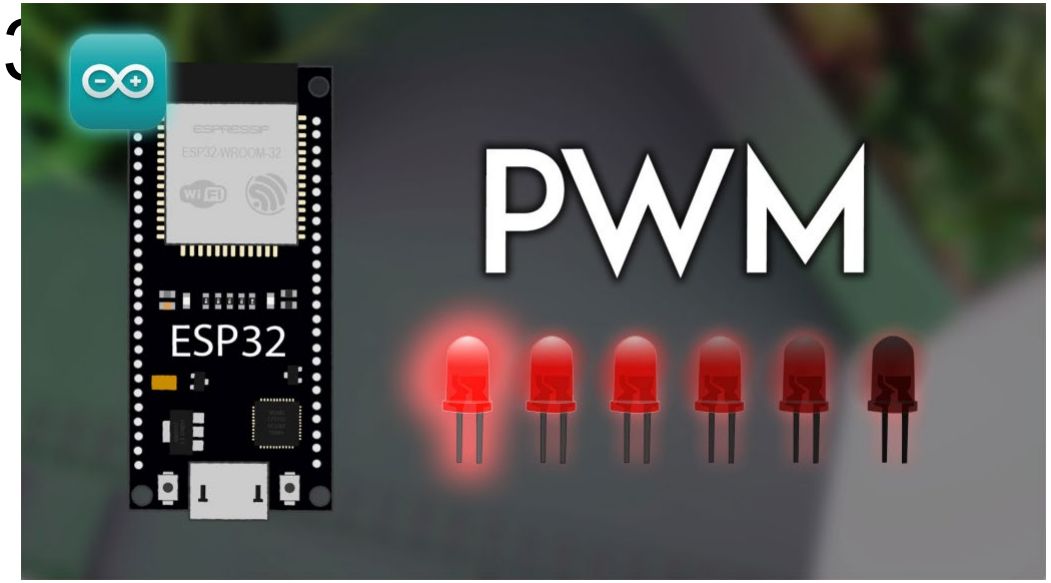
- Then how can ESP32 measure varying voltage levels (AKA Analog Input)?
  - The answer: A sub-system called "ADC"
  - NodeMCU ESP32 has 2 pins capable of ADC.

- Why do you need to master Analog Input?
  - Because a lot of useful sensors send Analog signals.
    - E.g. Temperature, Humidity, Distance, Brightness etc.

Important Task 3:
Use this Tutorial and the components in your Kit to learn how to utilize the Potentiometer with your ESP32:
https://randomnerdtutorials.com/esp32-adc-analog-read-arduino-ide/

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Analog Output using NodeMCU ESP3



- Finally, just like for Analog Input, microcontrollers cannot natively generate signals with varying voltage levels (Analog Output).
- Their pins can either be ON or OFF, but they can use this ability to generate Analog Output.
- This function is called Pulse Width Modulation (PWM).
- PWM is used to control motors, move Servos, dim LEDs and many other functions.

Important Task 4:
Use this Tutorial and the components in your Kit to learn how to utilize PWM for LEDs (as example):
https://randomnerdtutorials.com/esp32-pwm-arduino-ide/

Takeaway for this Week

1. You must complete Tasks 1-4 in these slides this week to master:
    - Setting up the NodeMCU ESP32 with your Arduino IDE to program it
    - Know how to use LEDs and Buttons as Digital IO
    - Know how to use the Potentiometer as Analog Input
    - Know how to do basic PWM using LED as Analog Output

2. You will notice that the tutorial site (Random Nerd Tutorials) contain a list of 160+ tutorials for projects using the ESP32, including:
    - Guides on how to use basic Digital and Analog Input/Output
    - Advanced functions like Timers, Interrupts, Webservers and More
    - Possible Project Ideas

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IoT Track – Week 3

Getting Good with Arduino Programming

Prepared by Dr. Mark Tee Kit Tsun (SUTS)

The Arduino Programming Language

- "Arduino" is actually the name of the whole Electronics Development Ecosystem. It consists of:
  A. The Arduino Board consisting of a microcontroller running the Arduino Firmware
  B. The Arduino IDE (a free programming software to write scripts for the Firmware to run)
  C. The Arduino Language



- This week's focus is on helping you master the Arduino Language.
  - The primary syntax is based on C++ but is not exactly C++.
  - Has built-in libraries to operate common complex components.
  - You can download and access additional libraries for many other components.
  - A lot of non-Arduino microcontrollers have support for this language, but you must download and include the board type into the IDE first.

- Begin by familiarizing yourself with where to find help:
  1. The official Website - https://www.arduino.cc/ (find, download & Install the Arduino IDE)
  2. Reference for standard Arduino functions & libraries - https://www.arduino.cc/reference/en/
  3. Learn the language here - https://www.arduino.cc/en/Tutorial/HomePage

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Level 1: C Programming in Arduino (2 Hours)

Just like learning a new language, you must methodically read and try the examples in this order:

1. Program Structure of a Sketch (the Arduino source code file)
2. Data Types available
3. How to set Variables
4. Using Operators
5. Decisions using If..Else and Switch
6. Loops using for, while and do
7. Writing Functions
8. Using Strings and String Objects
9. Control Timing
10. Using Arrays



Caution!

Even if you are familiar with C/C++, you must go through these tutorials and try them out on Arduino IDE at least once.

Level 2: Revisiting ESP32 Electronic Functions (1 hour)

Now that you are proficient in the C Syntax, review the tutorial examples in Week 2 to revise on how these functions work:

1. pinMode() – used in the setup() function to configure a pin to be INPUT or OUTPUT
2. digitalRead() – used to read an ON/OFF signal from a sensor.
3. digitalWrite() – used to send a ON/OFF signal to the actuator.
4. analogRead() – used to read a variable-voltage level signal from an Analog Pin.
5. analogWrite() – used to output a variable-voltage level signal to a PWM Pin.
6. Serial.begin() – used in the setup() function to start the Serial Communications feature.
7. Serial.print() – used to transmit a string through the COM line towards your PC.
8. Serial.println() – same as print(), but this one will end with a new line.

The full Reference can be found here: https://www.arduino.cc/reference/en/

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Level 3: Look for Examples

Depending on what you want to achieve in your ESP32 (or Arduino Uno), you can look for examples in primarily 2 places:

1. Built-in Examples
    - In Arduino Uno, go to File > Examples; browse and find an example file that can show you how to do a specific function or use a feature.
    - Explanation for each example can be found here: https://docs.arduino.cc/built-in-examples/

2. Tutorials Site and Example Uses
    - ESP32-related How Tos: https://randomnerdtutorials.com/getting-started-with-esp32/
    - Look for example videos like https://www.youtube.com/watch?v=tc3Qnf79Ny8

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Takeaway for this Week

1. Once and for all, it is time for you to master the Arduino Programming Language. Dedicate a day to go through C programming and Arduino functions.

2. Review how the ESP32 works and how it differs from Arduino Uno.

3. Remember, your equipment includes an Arduino Uno, if you want to implement non-Internet applications.

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IoT Track – Week 4

No Slides, Just Action -> Part 4 Canvas Page

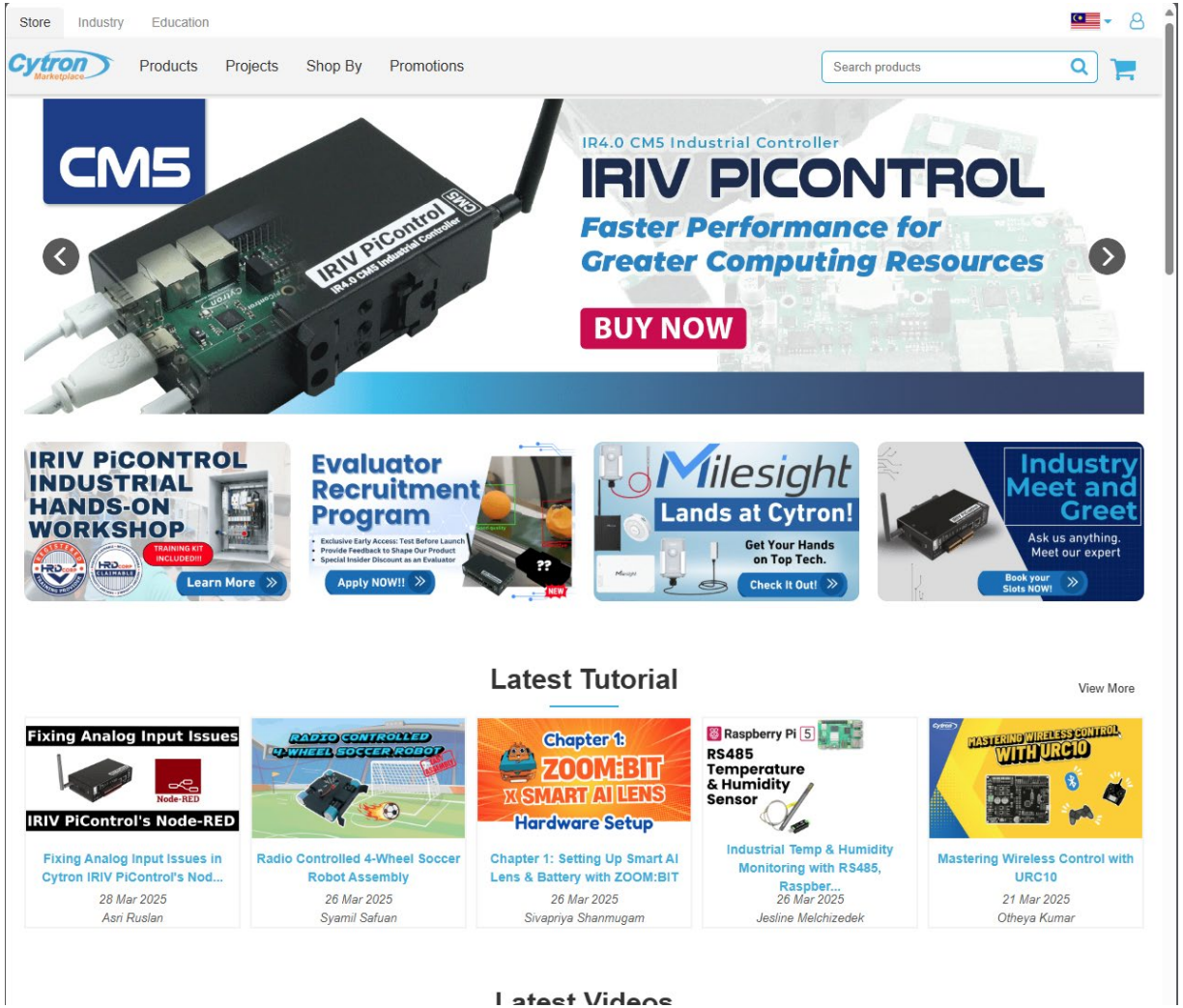
Prepared by Dr. Mark Tee Kit Tsun (SUTS)

What's Going On?

- At this point, you should already be familiar with what the ESP32 is and how to use it to interface with electronic sensors and actuators (digital/analog input/output).

- Action Time!
  - Explore the Market for possible components – Cytron.io
  - Create a Student Account for Free Shipping
  - Learn to Browse and Search for useful components
  - Find **Component Specifications**, Understanding Them, and Look for **Usage Instructions.**

- Notes:
  - **Only buy if necessary**
  - **Consider Safety First**
  - **Allocate Time for Shipping and Mistakes in your Project Plan**



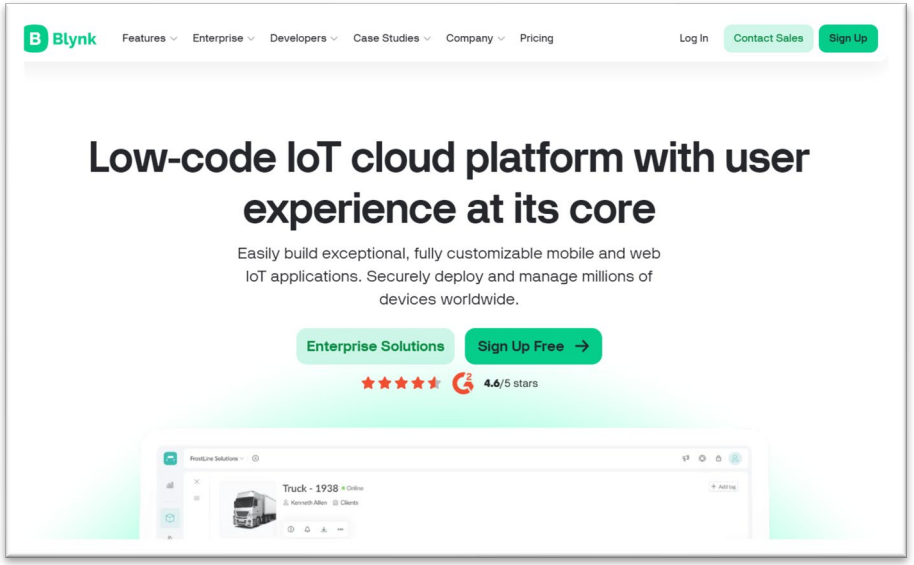
IoT Track – Week 5

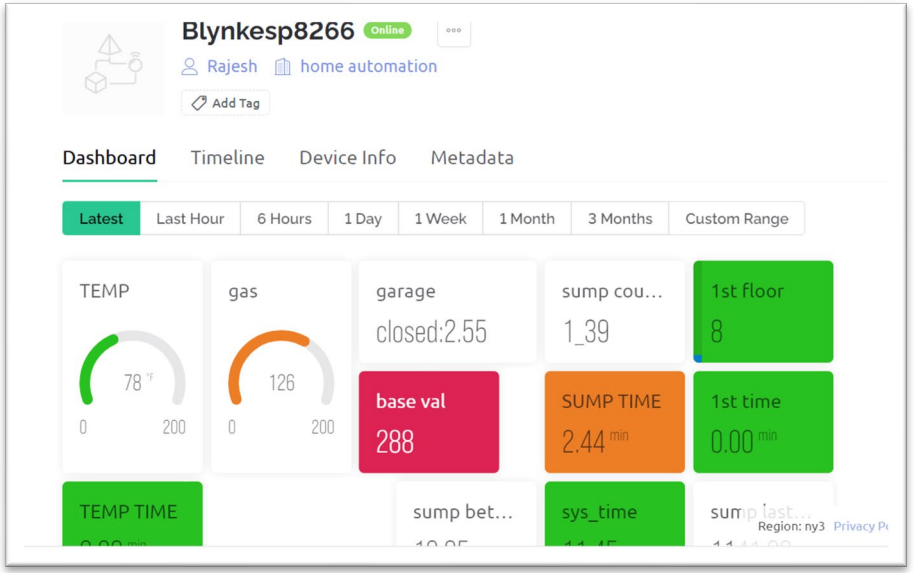
Using Blynk with NodeMCU ESP32

Prepared by Dr. Mark Tee Kit Tsun (SUTS)

Why Blynk ?

- Blynk is an Online Platform that provides a convenient Web Dashboard for your ESP32-based IoT machine.
- It has both web and mobile app interfaces so you can display buttons, labels, graphs and other widgets for displaying data from your ESP32, or trigger functions to execute.
- It does not have any direct option for interacting with other external services (e.g. database, virtual machines, AI, social media, etc.).
- Although Blynk has some limited machine-to-machine communication options (Automation), they require paid subscription to unlock.
- Still, it is useful for quickly testing your IoT system and demonstrating its capabilities.
- To know more, visit the official website: https://blynk.io/



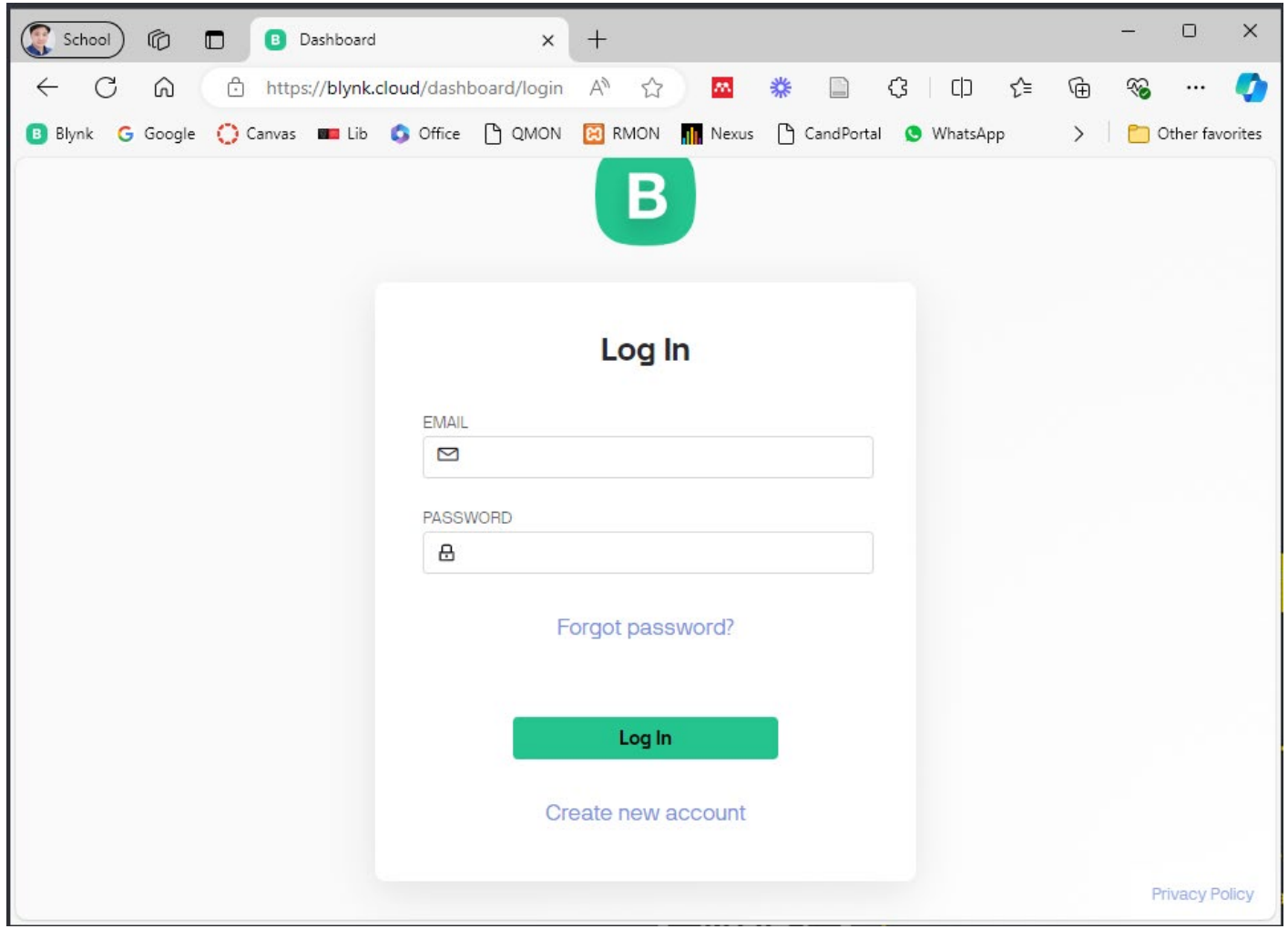


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First, Create Your Free Account (If you don't have one)

- Go to https://blynk.cloud/dashboard/login
- Create a New Account and Activate it using your Email.
- Log into Blynk.
- You will see your Blynk.Console

The Blynk.Console is the hub where you manage your projects (Developer Zone), connected Devices, their Users and even your Organisation/Location details.

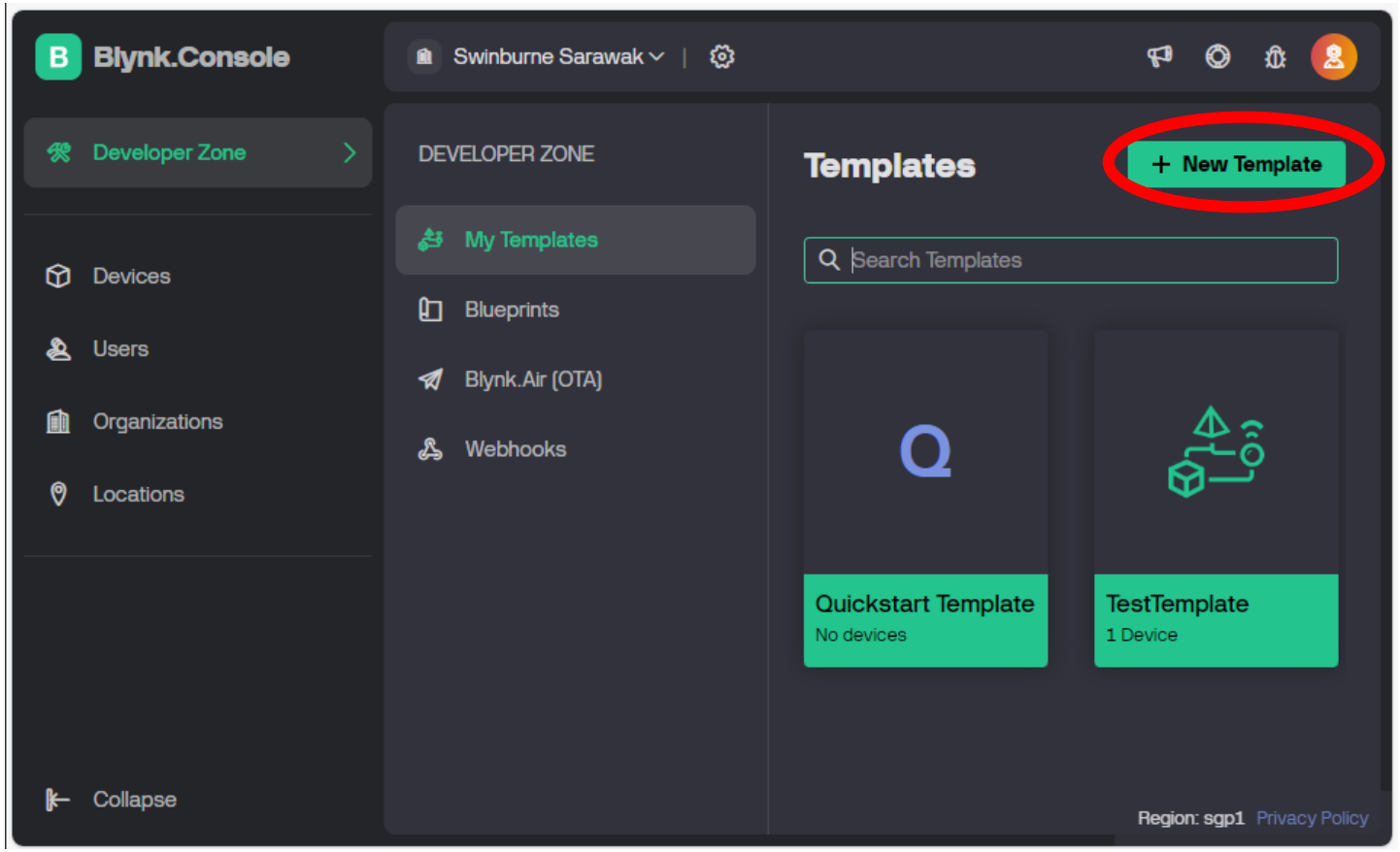


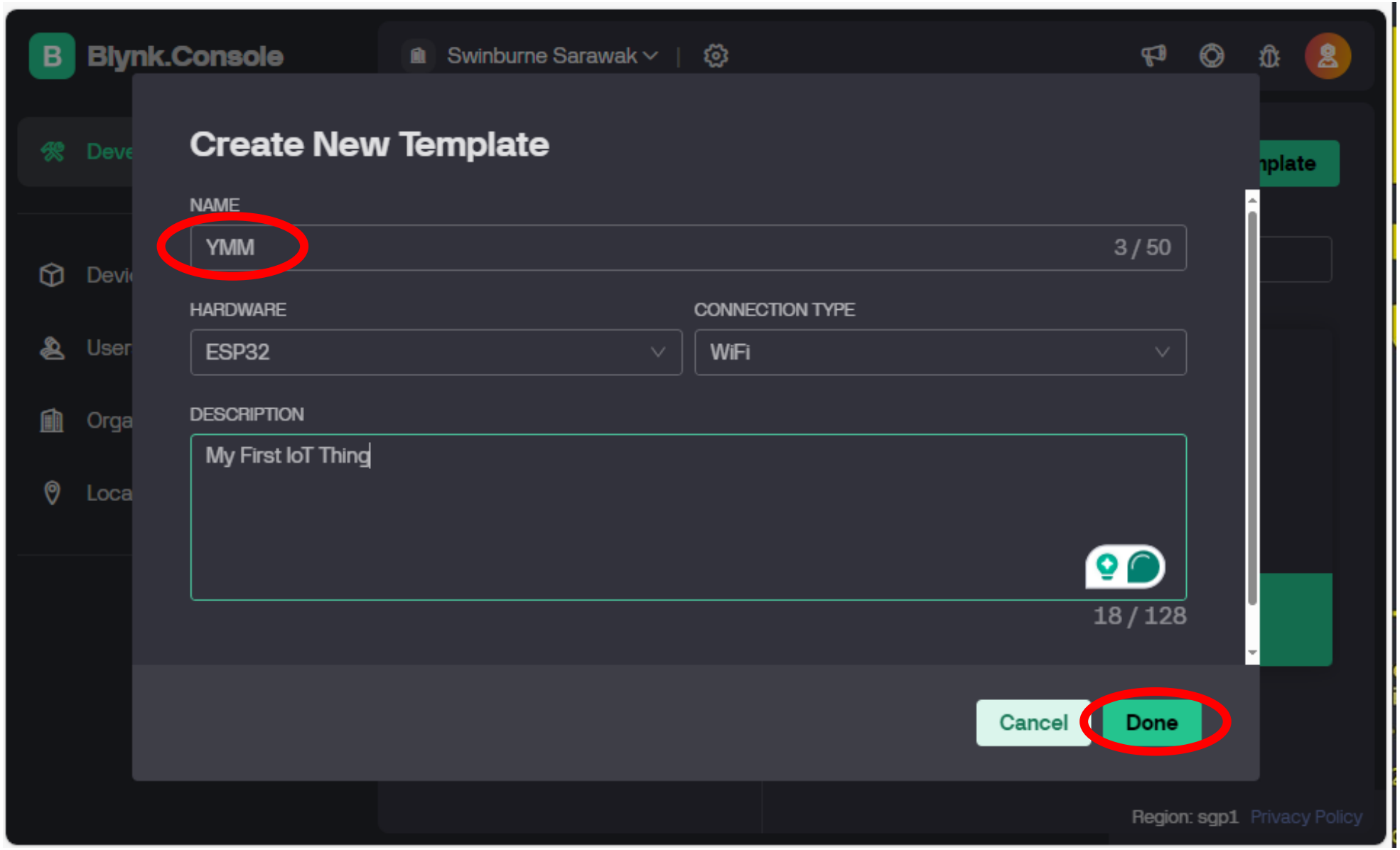
Create Your 1st Template

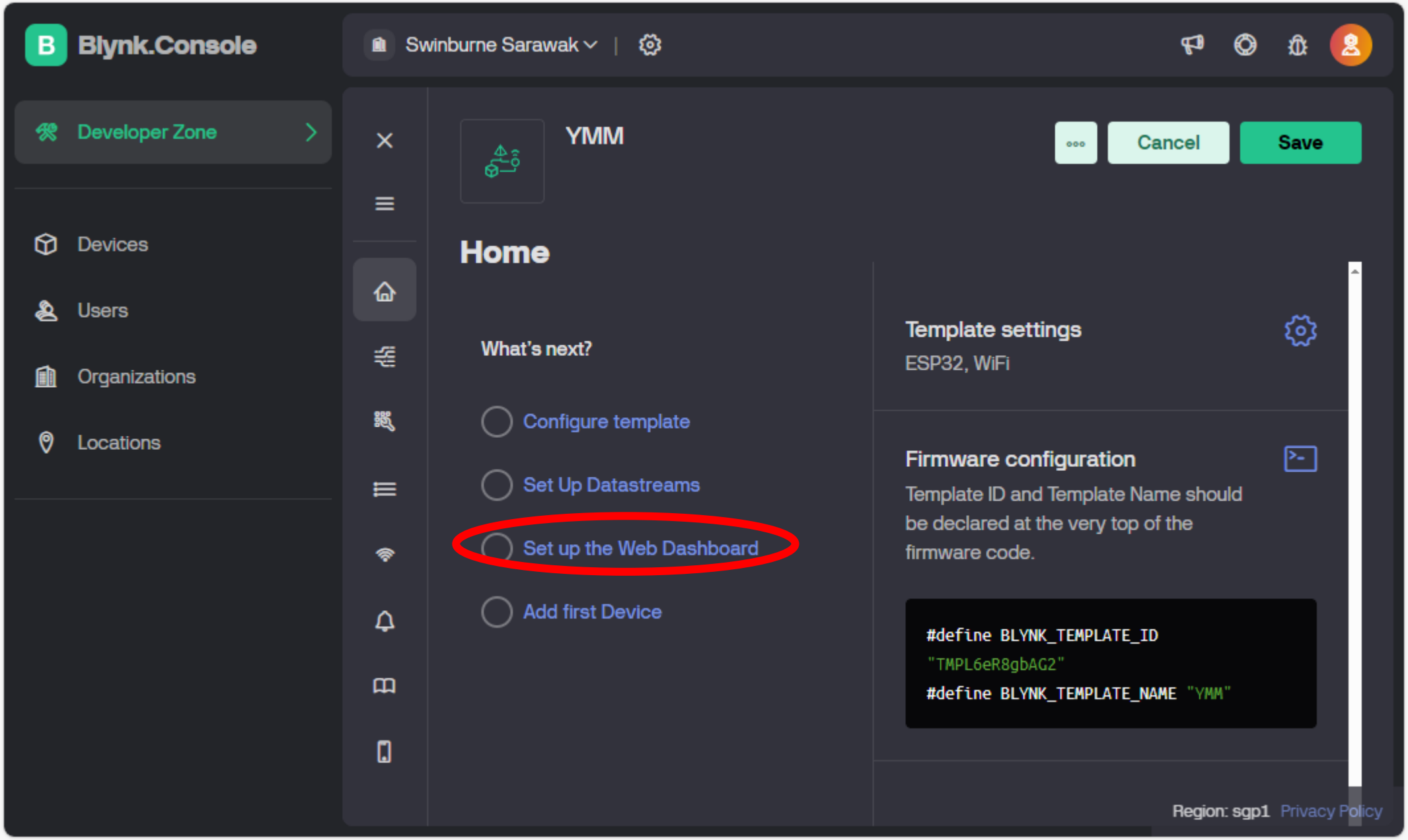
- Go to “Developer Zone” and Choose “+ New Template”.

A Template is an IoT Project. It holds information such as the Datastreams, Dashboard Layout and other project information.

Once a Template is built, you can add multiple ESP32 devices that can be controlled using the Template’s contents.

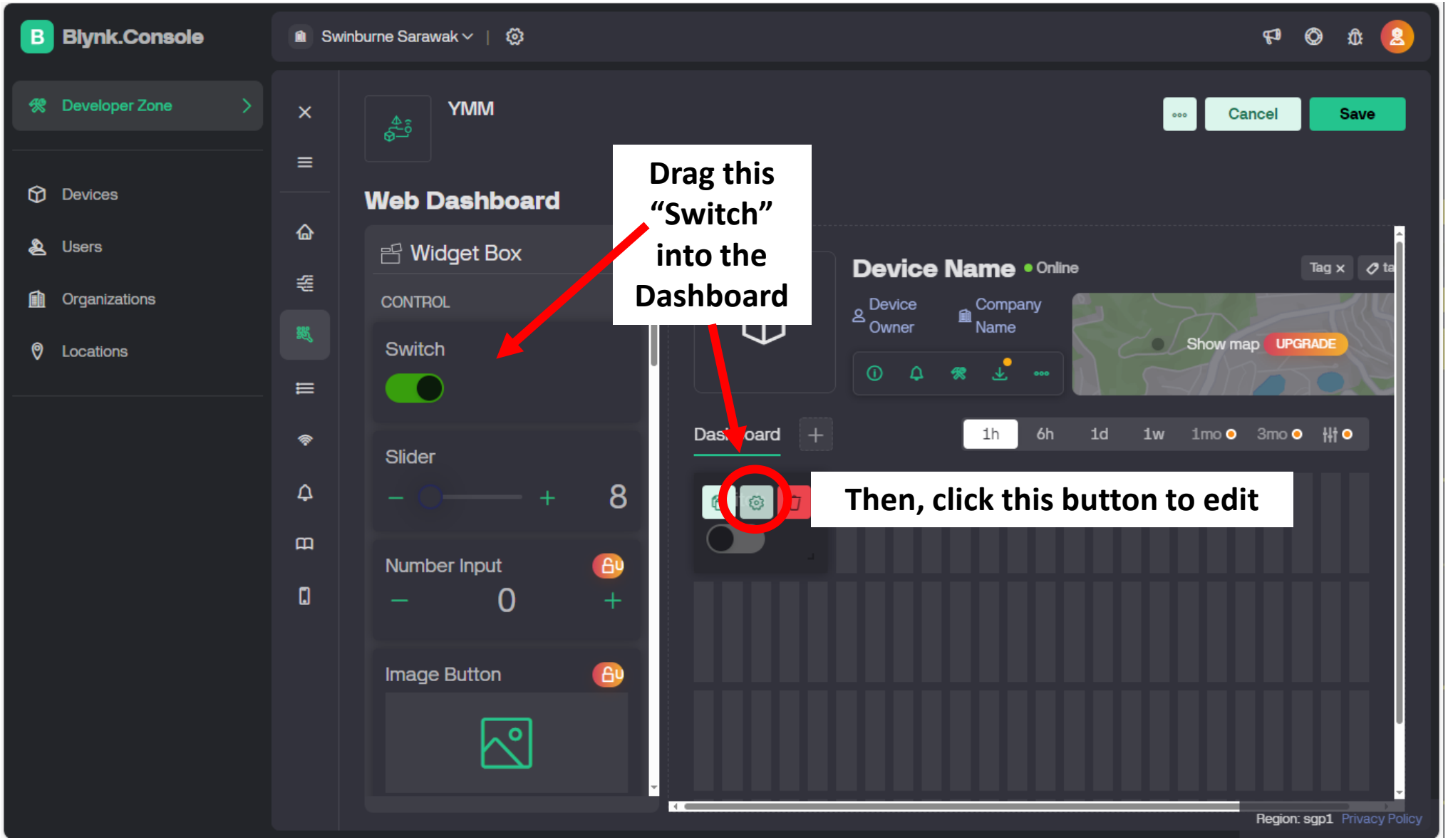



Blynk.Console
Swinburne Sarawak
Create New Template
NAME
YMM
3 / 50
HARDWARE
ESP32
CONNECTION TYPE
WiFi
DESCRIPTION
My First IoT Thing
18 / 128
Cancel
Done
Region: sgp1
Privacy Policy

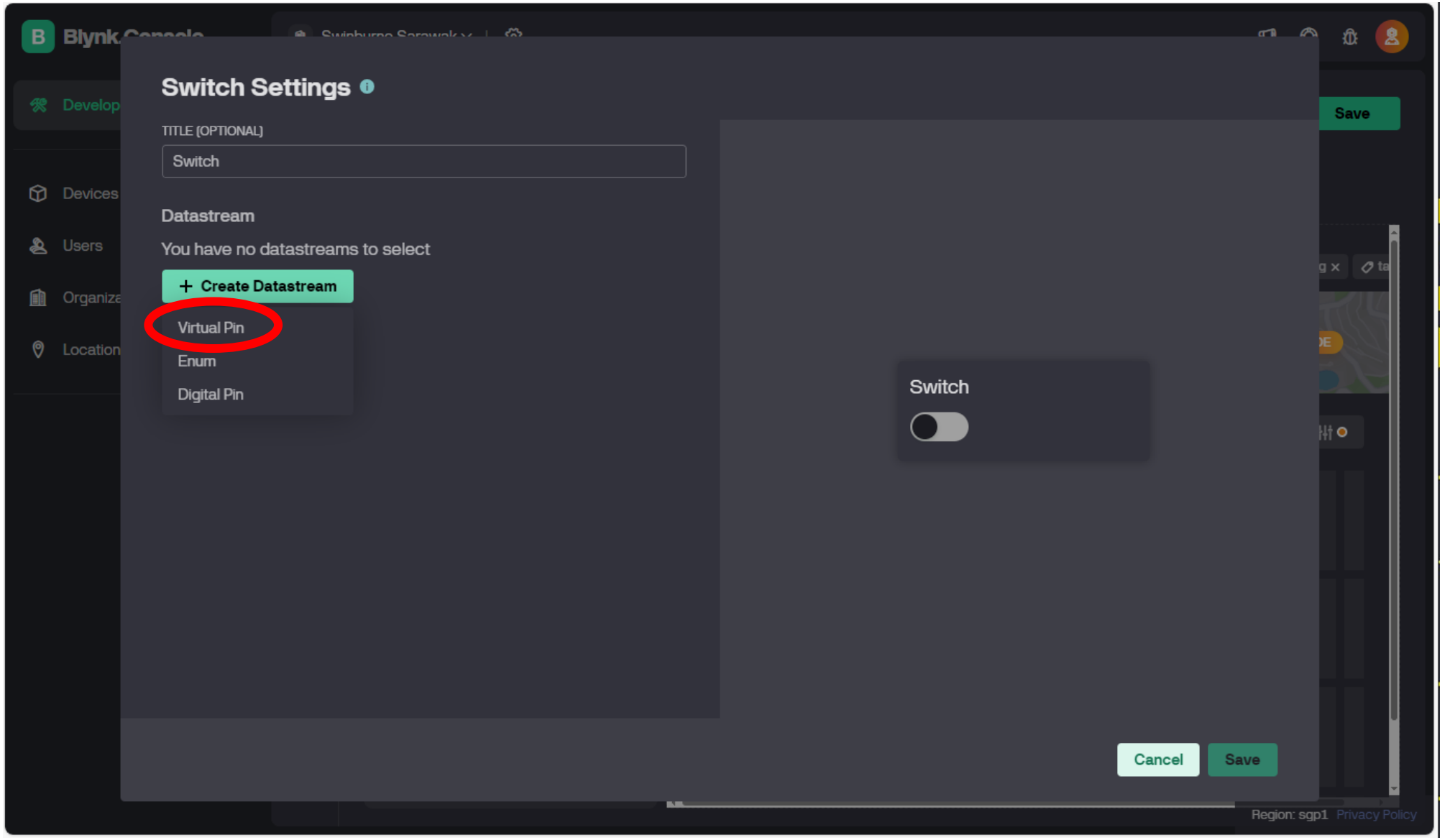


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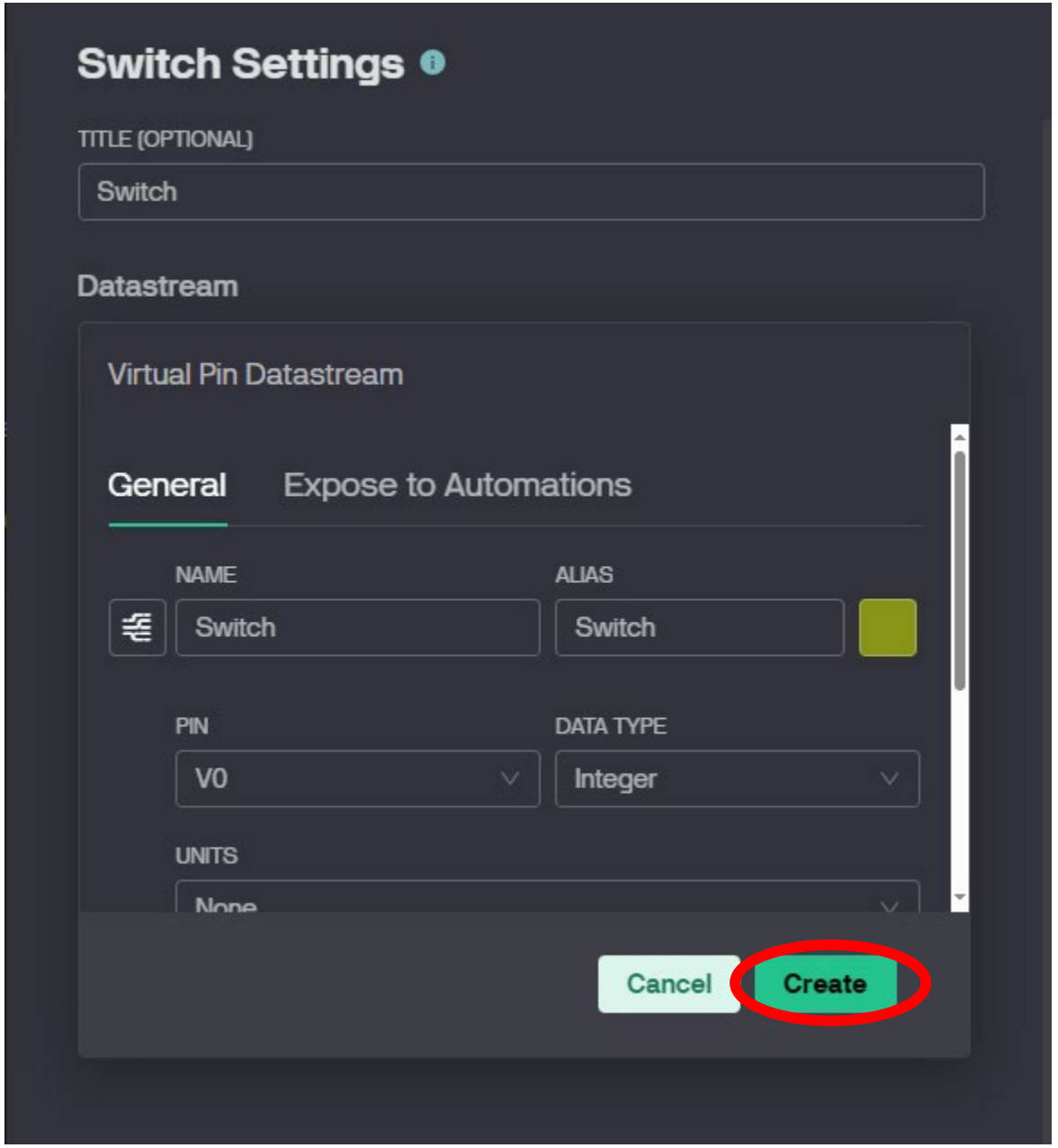
- Here, you can choose the type of widget you need, drag and drop it into the Dashboard, then click the Settings button to configure it.



- Choose to attach this control widget to a new Datastream, and select “Virtual Pin”.
- We use Virtual Pins to interact between the Dashboard and your ESP32.

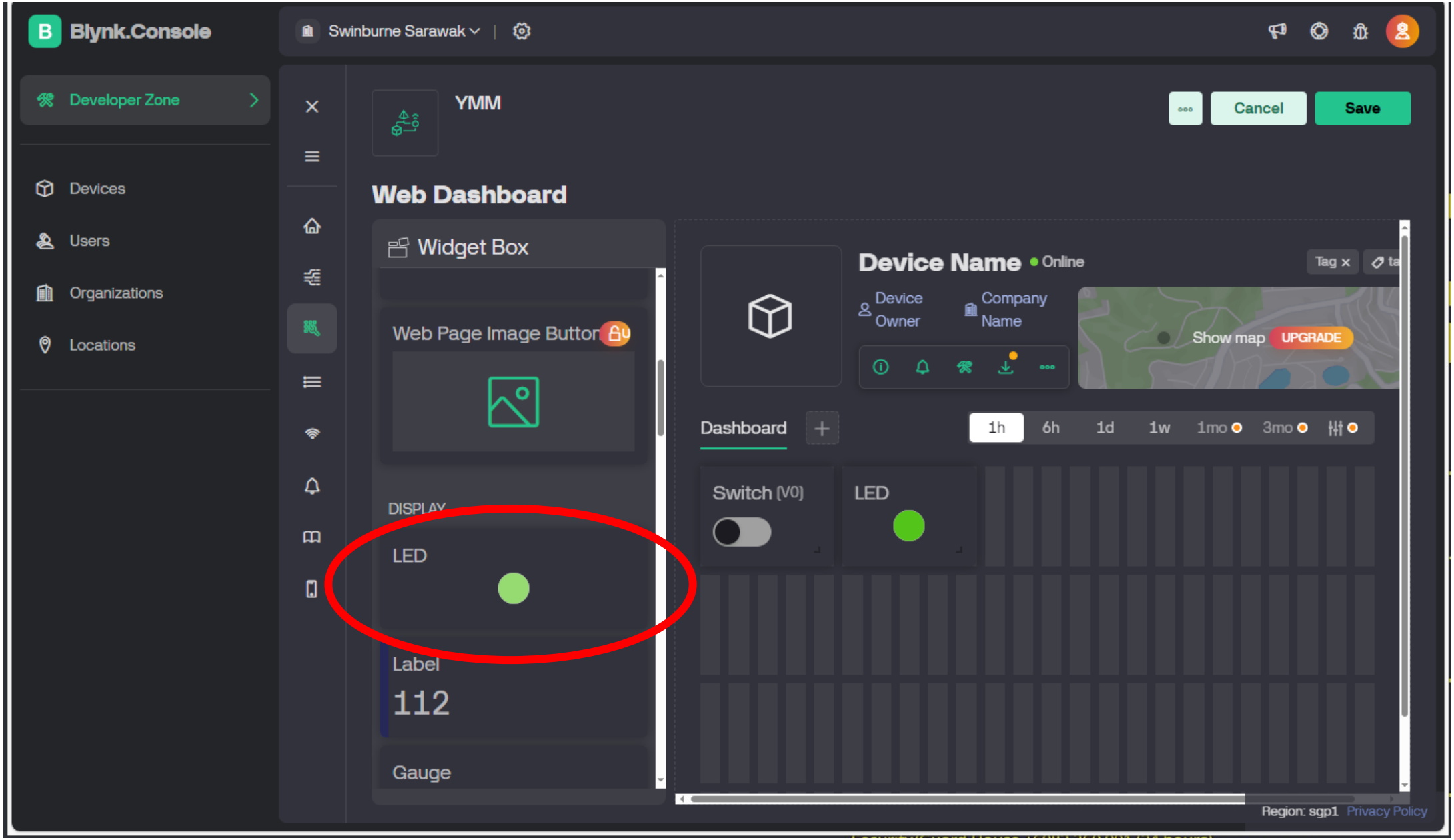


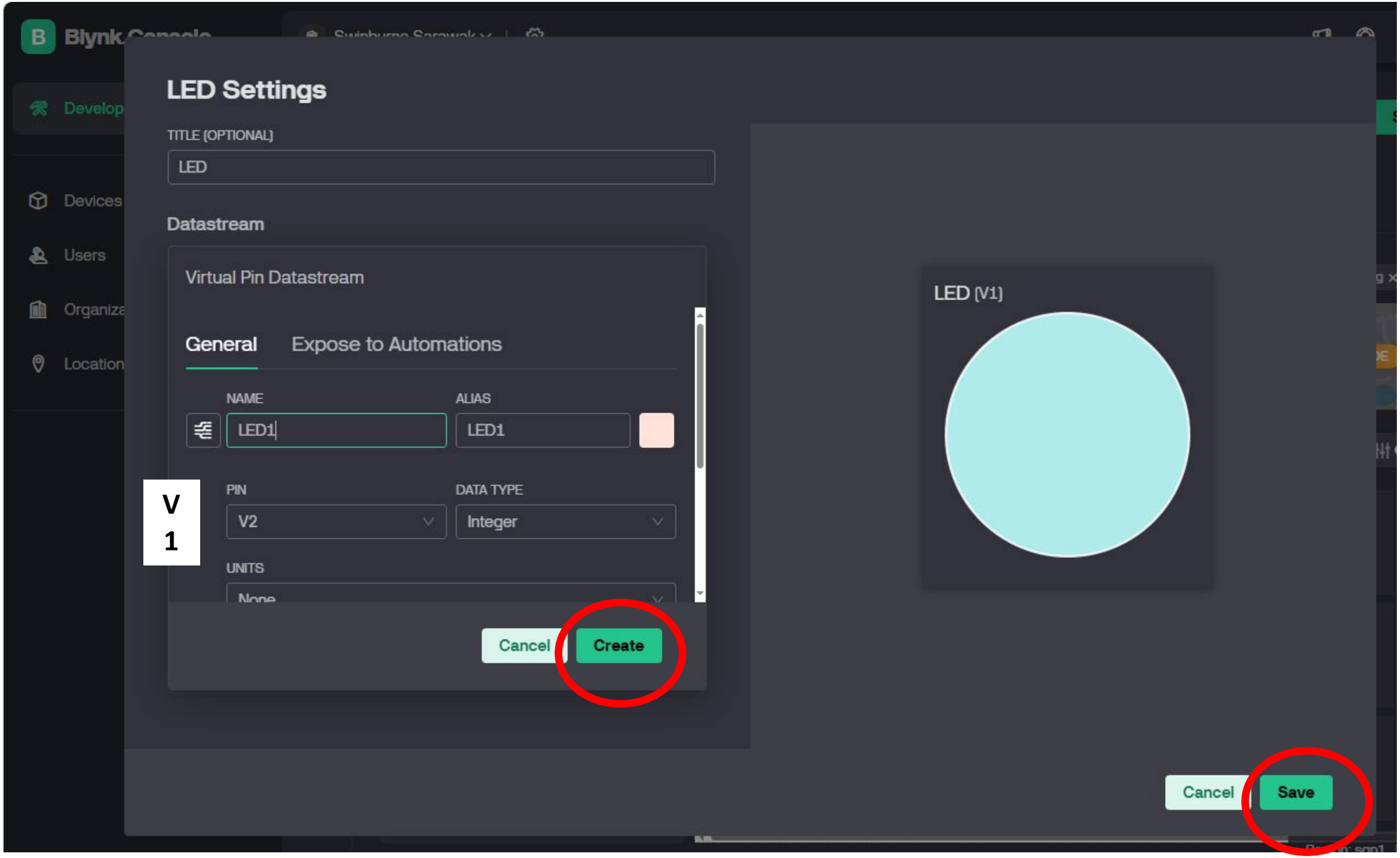
- You can configure details in the Virtual Pin.
- Pay close attention to which Pin this is (V0, V1, V2, V3, or V4).
- The Free account limits the choice to only 5 Datastreams.
- When done, press “Create”.
- You will now return to the Dashboard Editor to add other widgets.



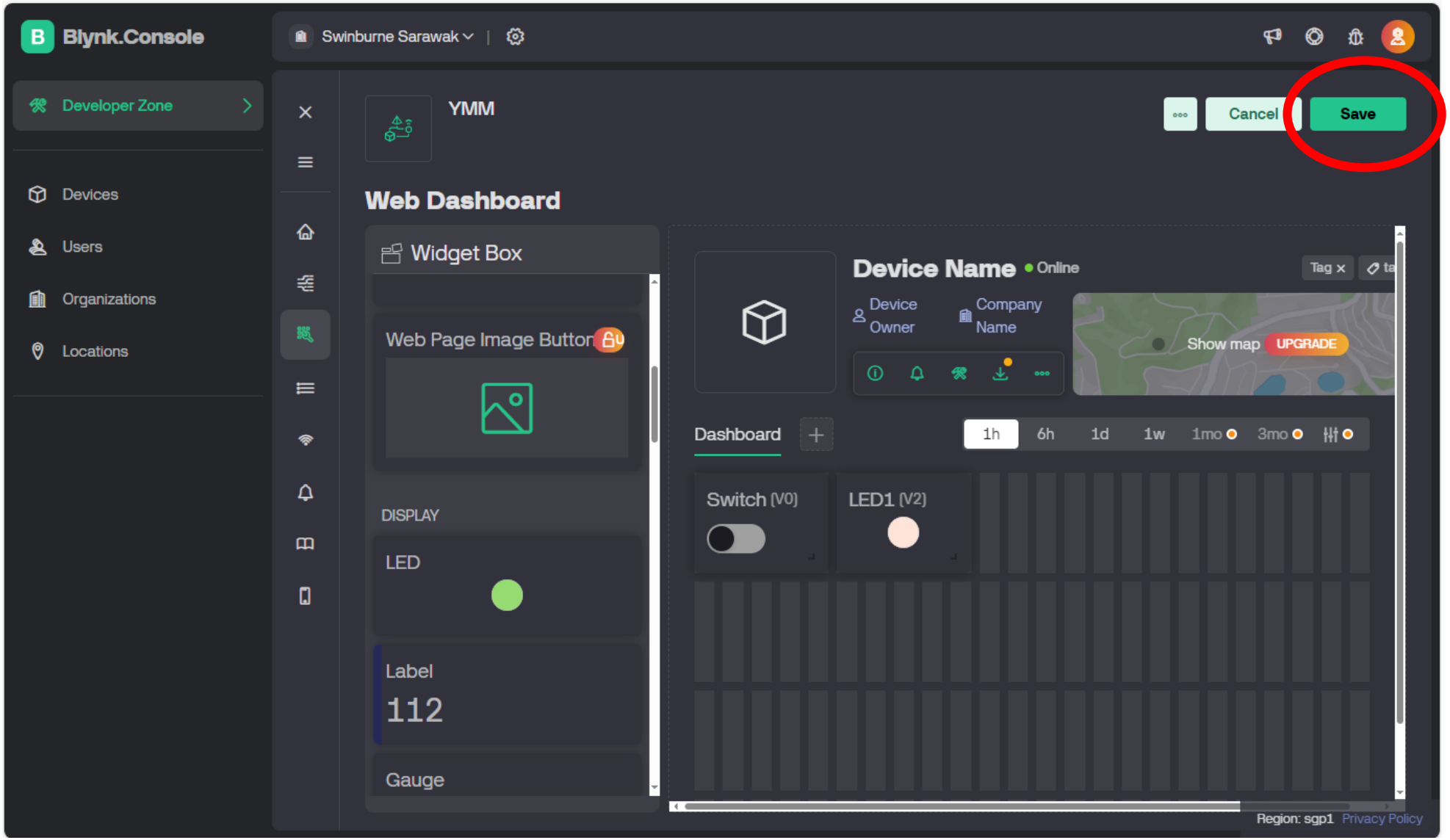
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- Adding another Widget now to represent Digital Output (LED) on the Dashboard.

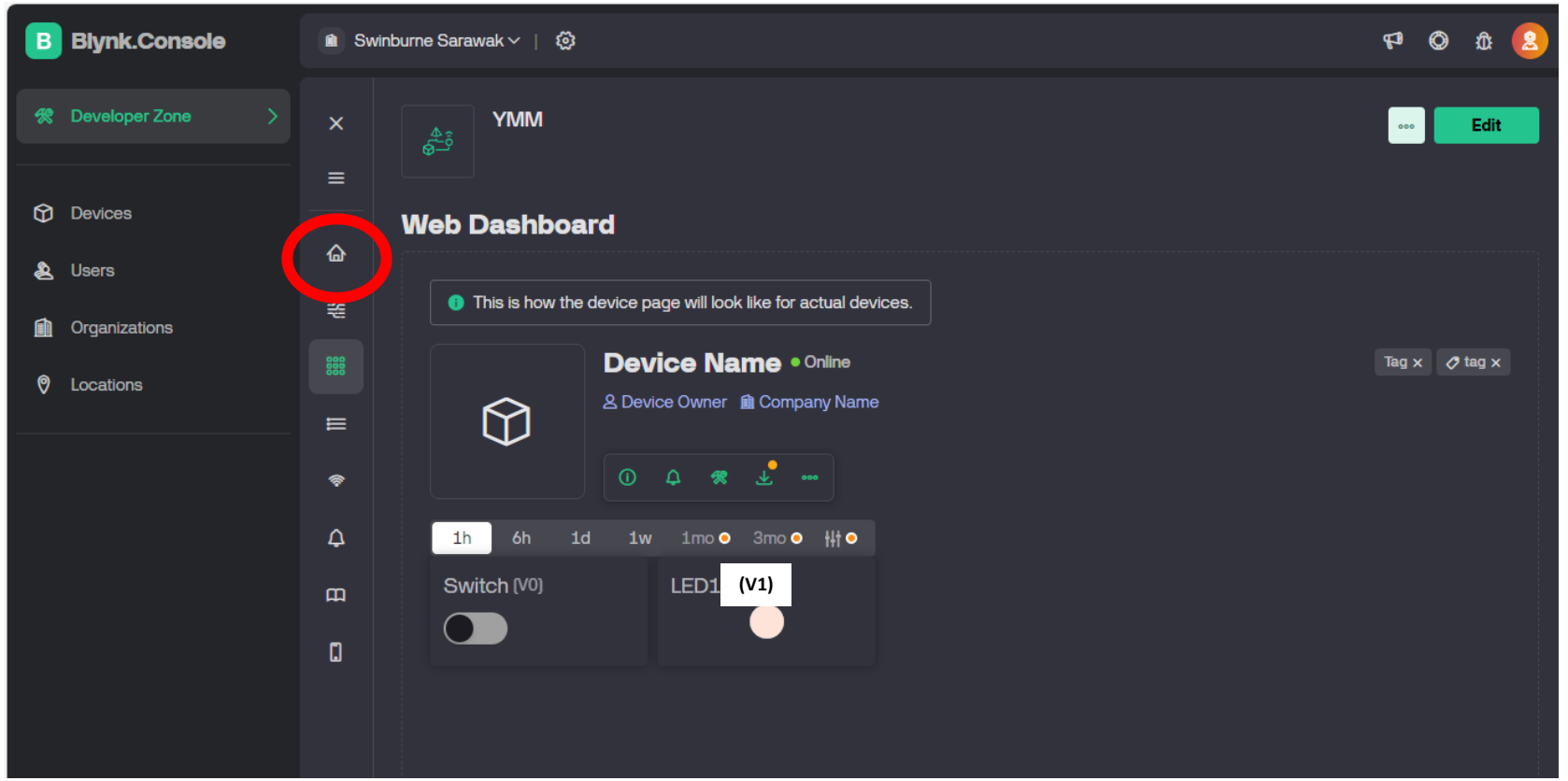



LED Settings
TITLE (OPTIONAL)
LED
Datastream
Virtual Pin Datastream
General
Expose to Automations
NAME
ALIAS
LED1
LED1
PIN
DATA TYPE
V2
Integer
UNITS
None
Cancel
Create
V1
LED (V1)
Cancel
Save

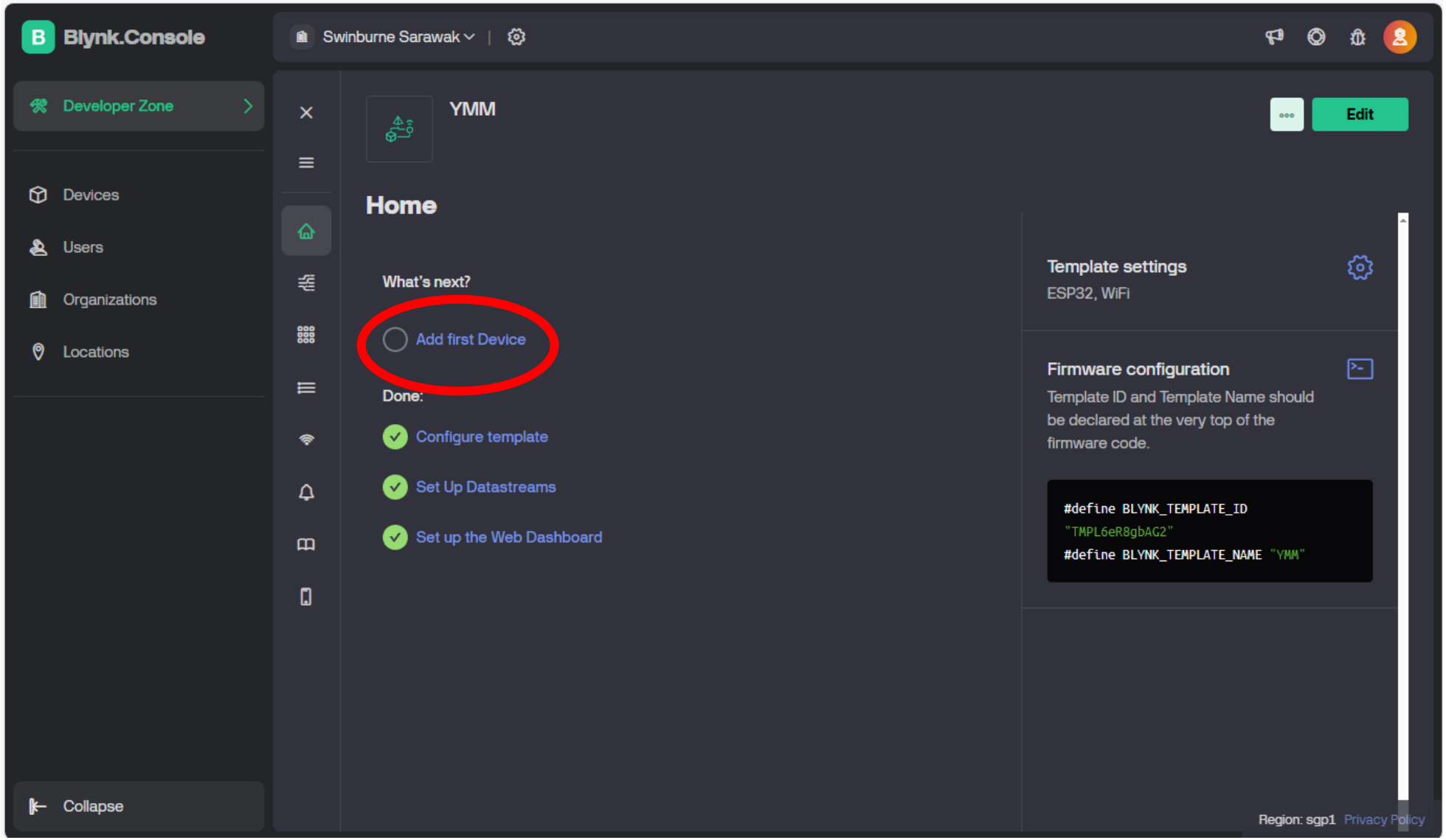
- After you are done, click “Save” to save work on the Dashboard into the Template.



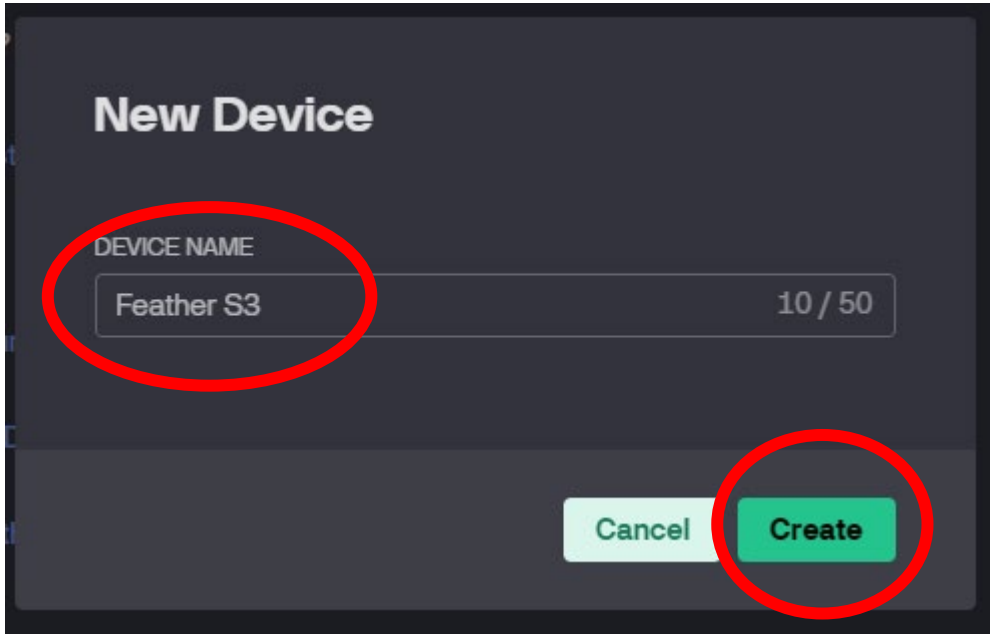
- If you have followed the previous steps, you have added 2 Control Widgets:
  A Switch (Virtual Pin V0), and a LED (Virtual Pin V1).
- Now, click this button to return to the Template Home Page.



- At the Template Home page, click “Add first Device”.
- Remember, it is possible for multiple ESP32 machines to be controlled through separate Blynk Dashboards.

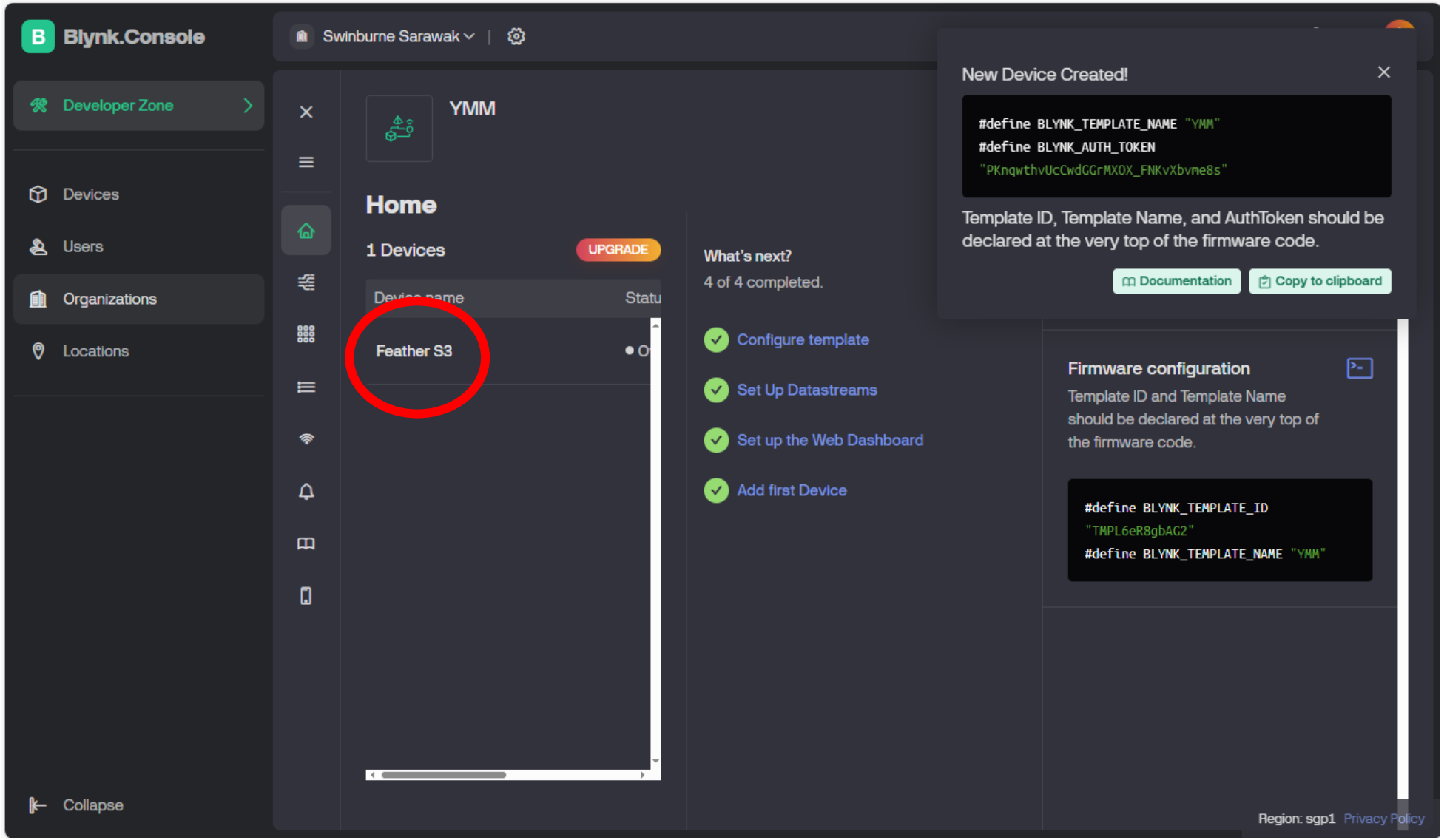


- This step is very important. Make sure to apply a unique name to identify your ESP32 machine.
- Once configured, this Blynk Device's Dashboard will be linked to your ESP32 board using a few lines of Identification Code.

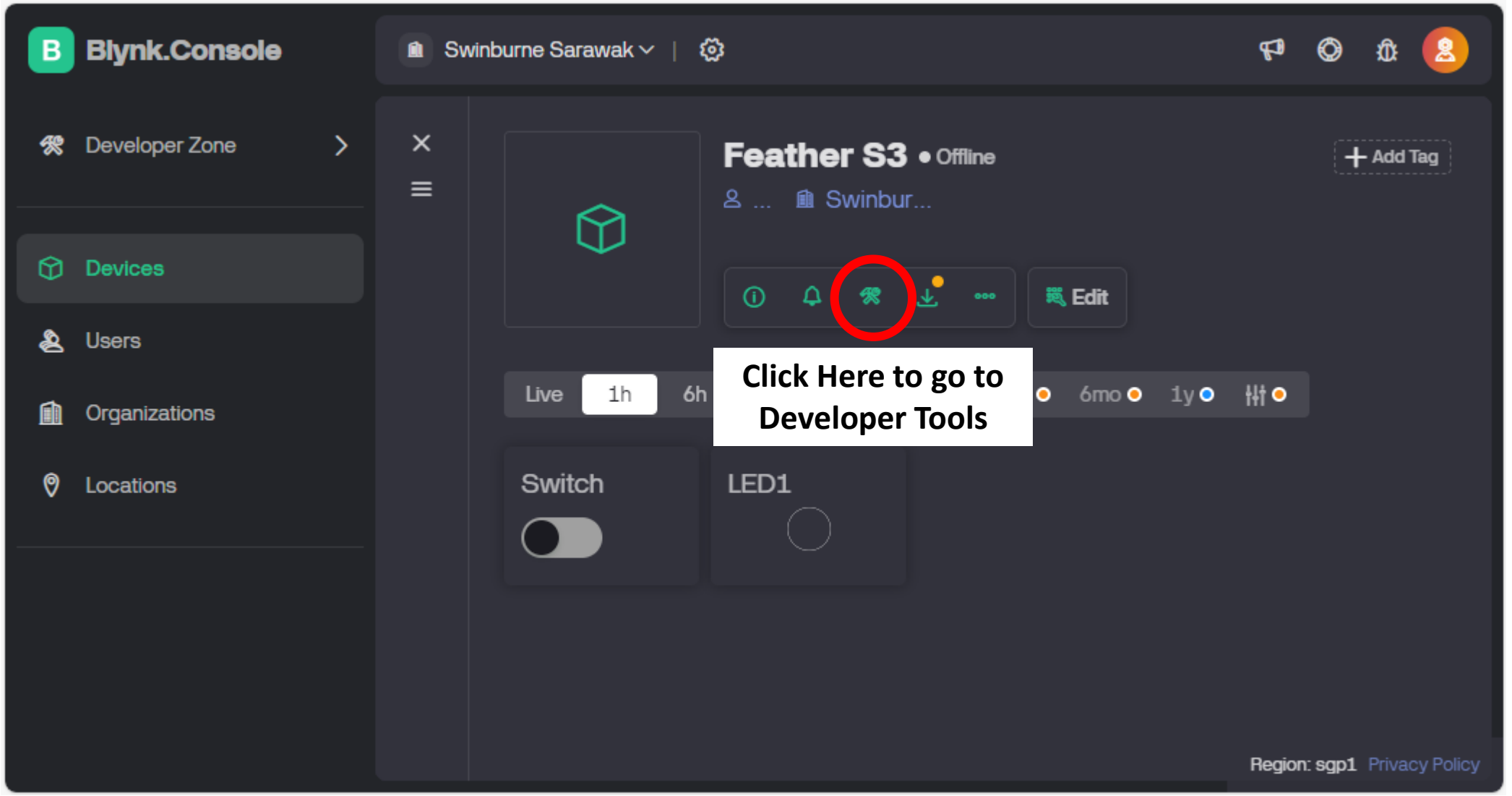


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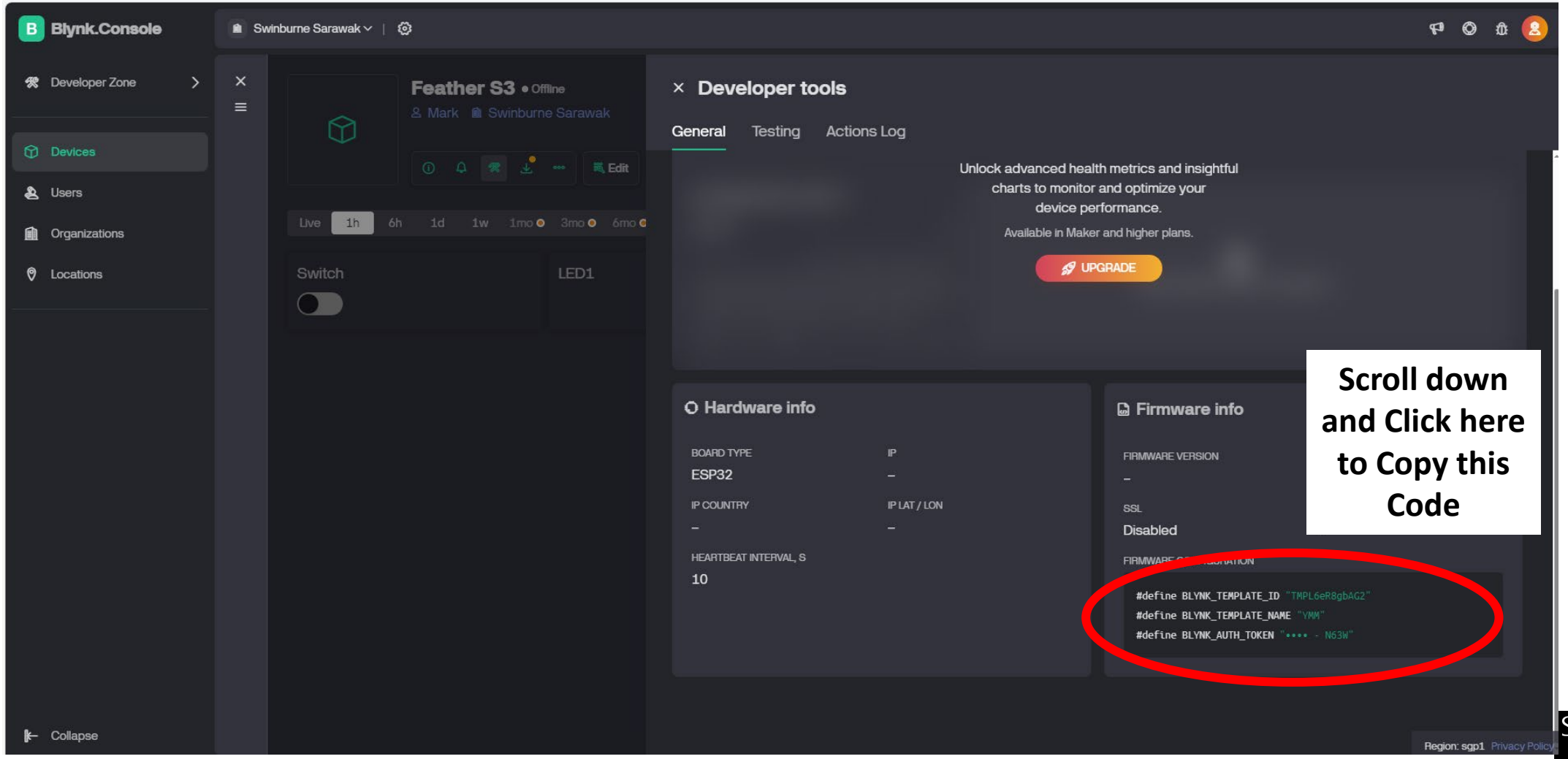
- Once created, click on the Device name to go to it's Dashboard.



- We need to copy that Identification Code. To do this, go to the “Developer Tools” in your Device’s Dashboard.



- In the Developer Tools panel, scroll down until you see the “FIRMWARE CONFIGURATION” section.
- Left-click it once to copy the Identification Code. We will need to paste this into our Arduino Sketch.



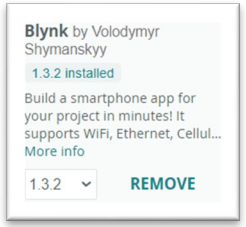
Then, Return to Arduino IDE and NodeMCU ESP32

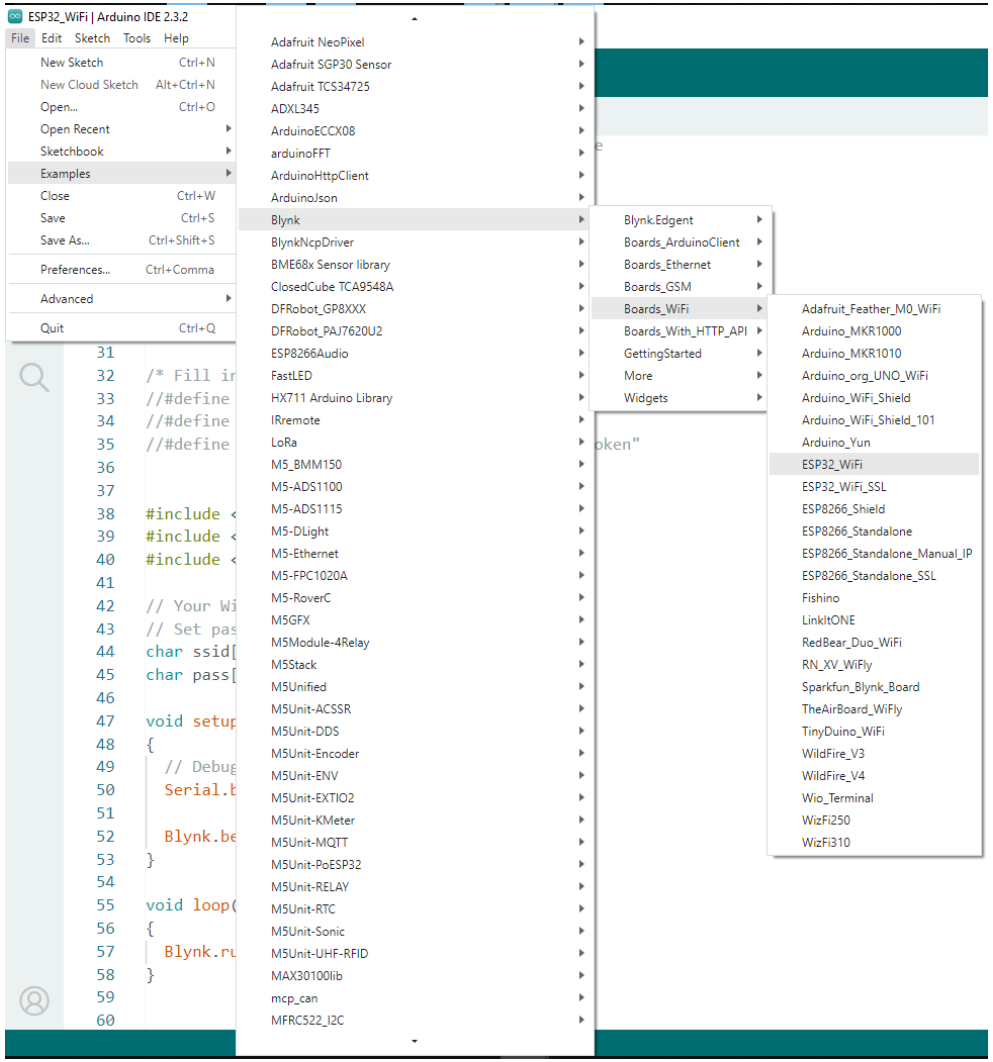
- Connect your NodeMCU ESP32 to the PC and configure your Arduino IDE to prepare programming it.
- Go to Sketch > Include Library > Manage Libraries… panel.
- Search “Blynk”. Install the latest version of Blynk by Volodymyr Shymanskyy.
- Go to File > Examples > Blynk > Boards_WIFI > ESP32_WIFI
- Save a new copy of this Sketch for your machine. (Don’t overwrite the example file).
- Paste the Identification Code from your Blynk into this part of the Sketch:

```
/* Fill in information from Blynk Device Info here */
//#define BLYNK_TEMPLATE_ID           "TMPxxxxxx"
//#define BLYNK_TEMPLATE_NAME         "Device"
//#define BLYNK_AUTH_TOKEN            "YourAuthToken"
```

- Don’t forget to input your WIFI for the ESP32 to connect to:

```
// Your WiFi credentials.
// Set password to "" for open networks.
char ssid[] = "YourNetworkName";
char pass[] = "YourPassword";
```

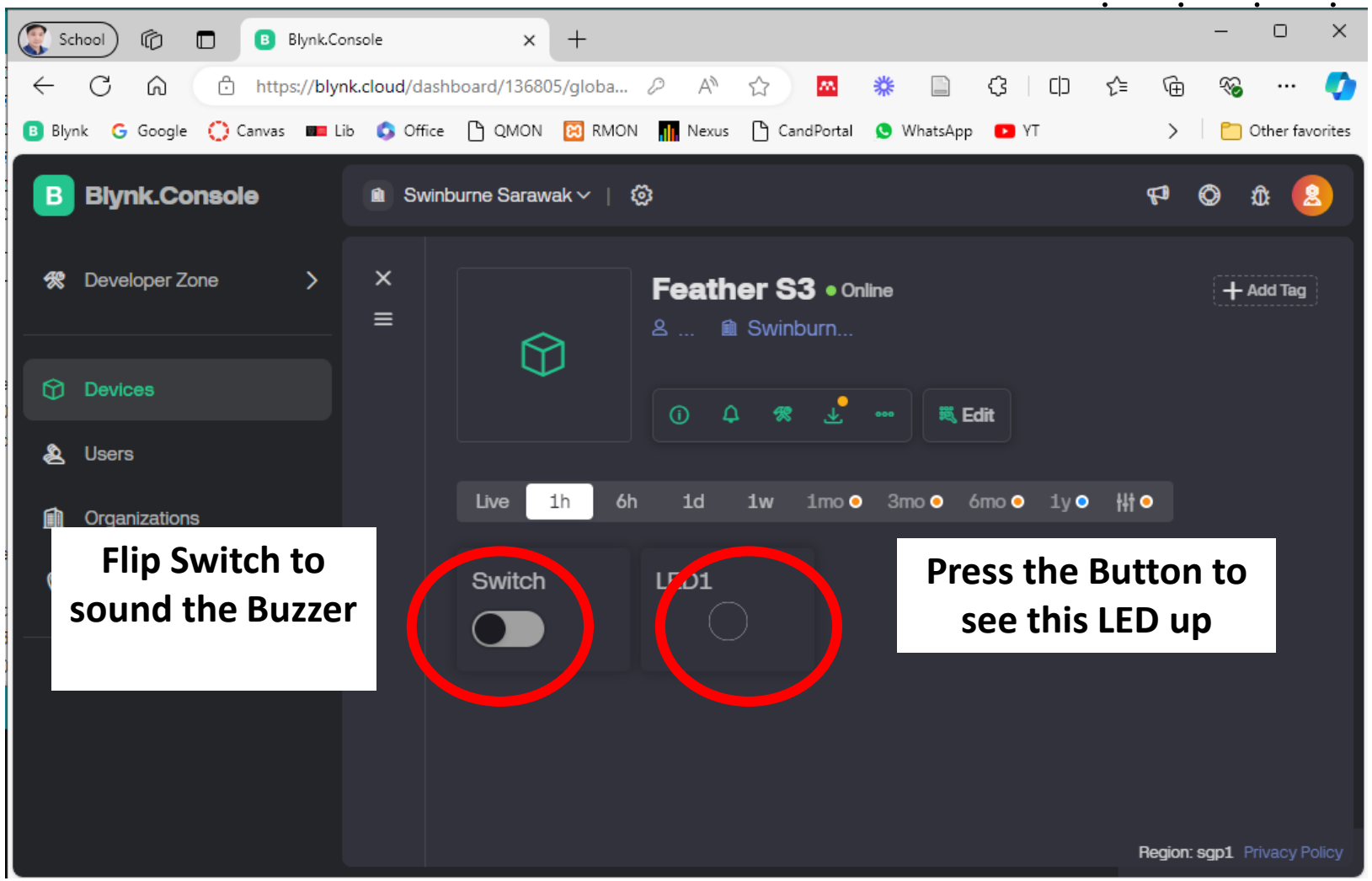




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- Refer to the other examples for getting V0 and V1 to interact with your ESP32.
- Find the example files here:

  File > Examples > Blynk > Getting Started
- The files:
  - GetData – Blynk_Write for updating from Blynk
  - PushData – Blynk TimerEvent for updating to Blynk



Takeaway for this Week

1. Blynk is useful for quick testing and controlling your ESP32 machines (projects wired to the ESP32 board.
2. You can use it for free but functionality is limited.
3. Refer to the latest official tutorial to learn how to use Blynk properly (these get updated quite frequently)

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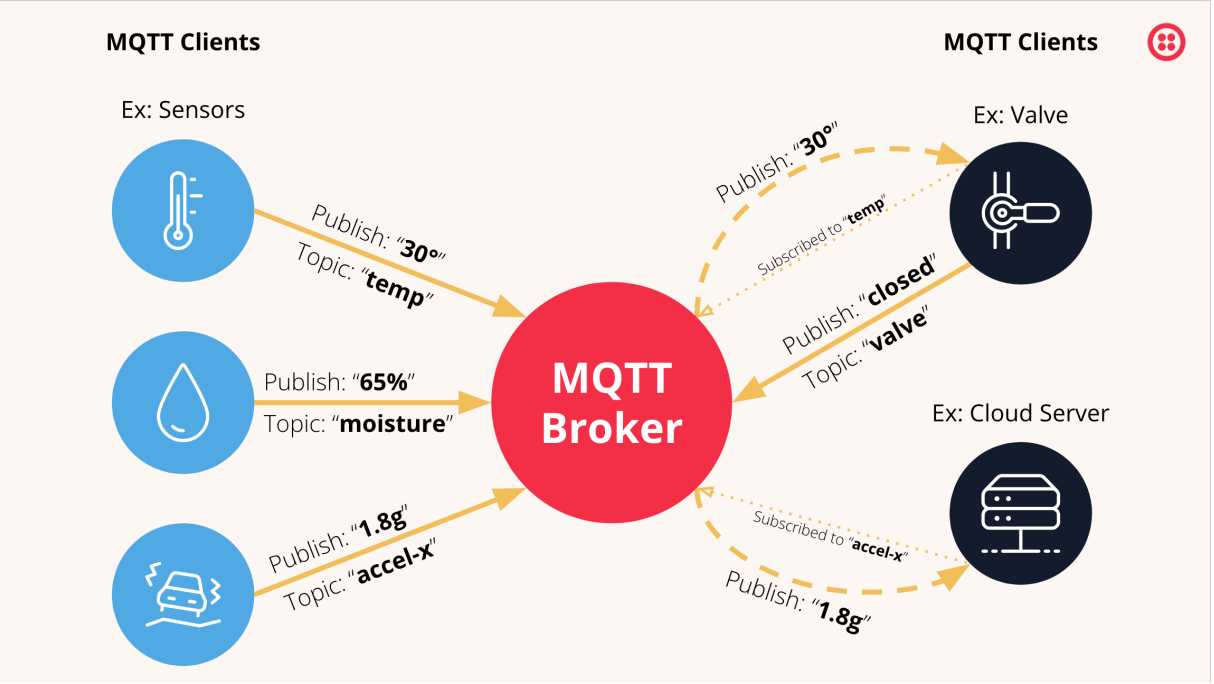
IoT Track – Week 6

IoT Communications with MQTT

Prepared by Dr. Mark Tee Kit Tsun (SUTS)

What is MQTT?

- Message Queuing Telemetry Transport (MQTT) is a data exchange Protocol (a way to pass messages).
- Especially used for Internet-of-Things applications because it is capable for massive Many-To-Many communications.
- Uses a Publish-Subscribe type of Client-Server Architecture.
  - Many receivers Subscribe to a Topic
  - When one sender publishes a message to that Topic, all those receivers get a copy of it



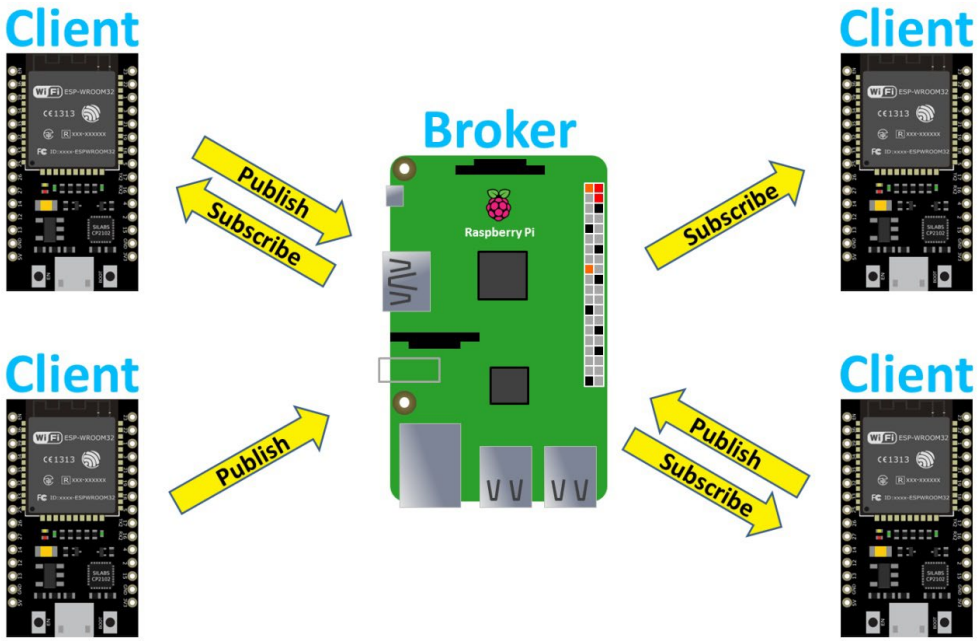


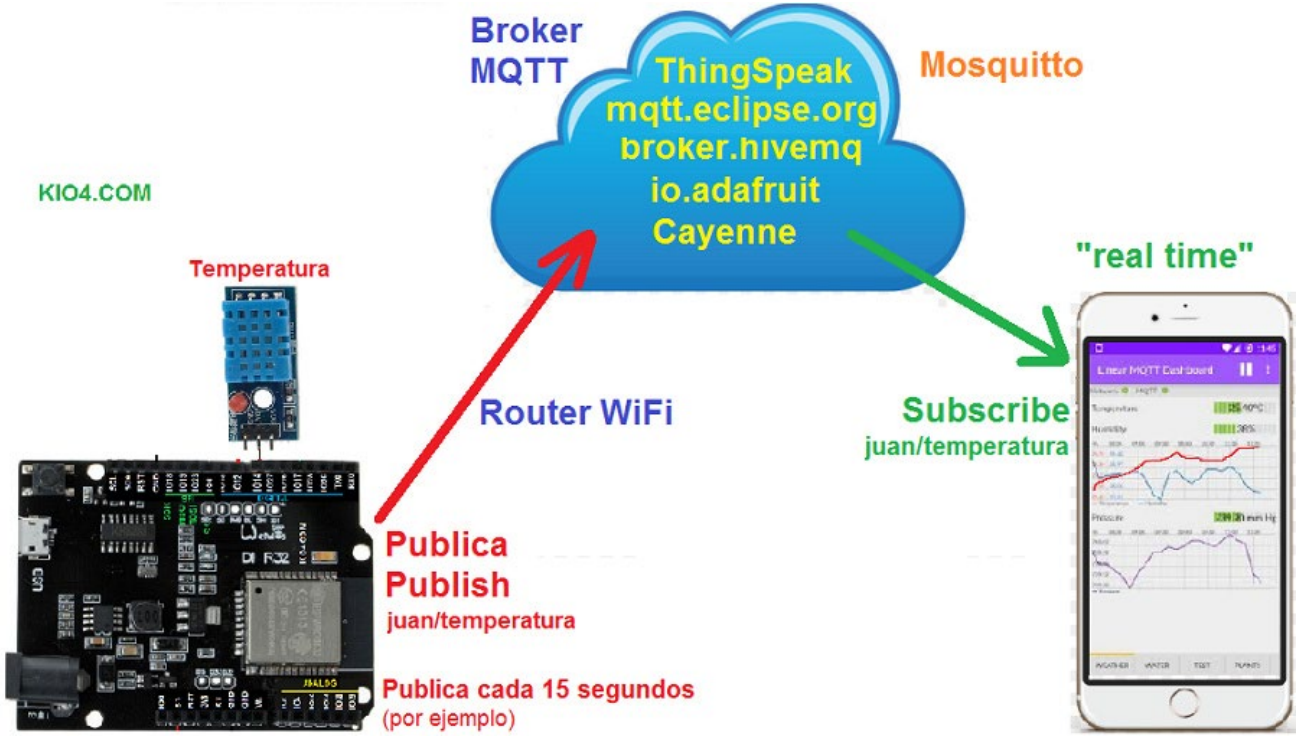
MQTT Basics: What is MQTT and How Does it Work? - YouTube

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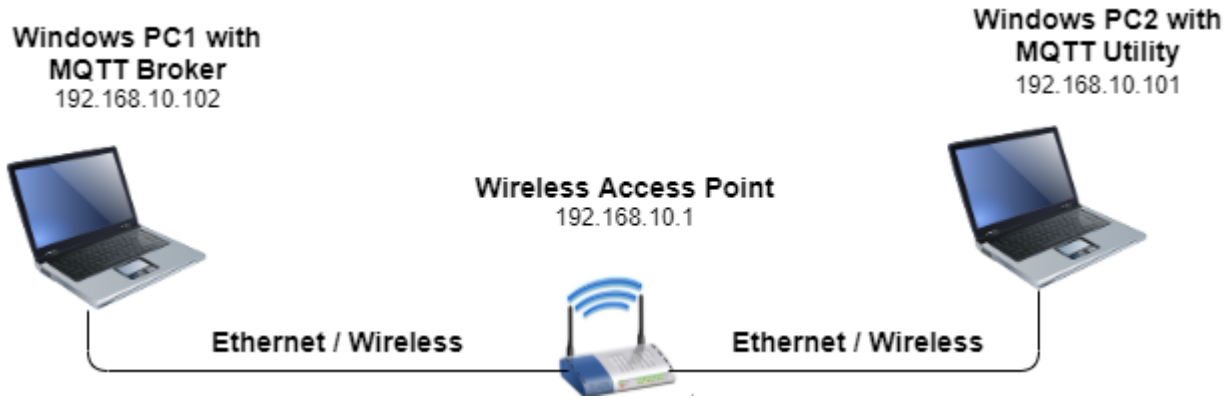
Why should I care?

- MQTT allows you to send and receive messages from multiple ESP32 machines, PCs, smartphones and more.
- You just need to download, install and configure an MQTT Broker in a computer on the same network.
- Each of the MQTT “Clients” must have a PubSubClient library to send and receive messages.
- There is an Open Source (AKA Free) MQTT Broker you can use, called Mosquitto.





Task 1: Setup a Mosquitto MQTT Broker on your PC



- The Mosquitto MQTT Broker is a server program that runs in the background of your computer.
- Follow the following tutorial to:
  - Download and Install Mosquitto on your Windows Computer
  - Configure the mosquitto.conf configuration file to allow anonymous mode and set your port to 1883.
  - Do Settings on your Firewall to allow Mosquitto to operate.

https://shop.theengs.io/blogs/news/installing-mosquitto-on-windows-and-make-it-accessible-from-your-local-network

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Task 2: Program the ESP32 to Connect to MQTT Broker

- You need to download and include the PubSubClient library in your Arduino IDE.
- Then, follow the tutorial to implement code in your NodeMCU ESP32 to practice sending and receiving messages via MQTT.
- You must have a working MQTT Broker running in a PC on the same WIFI first. Tutorial:

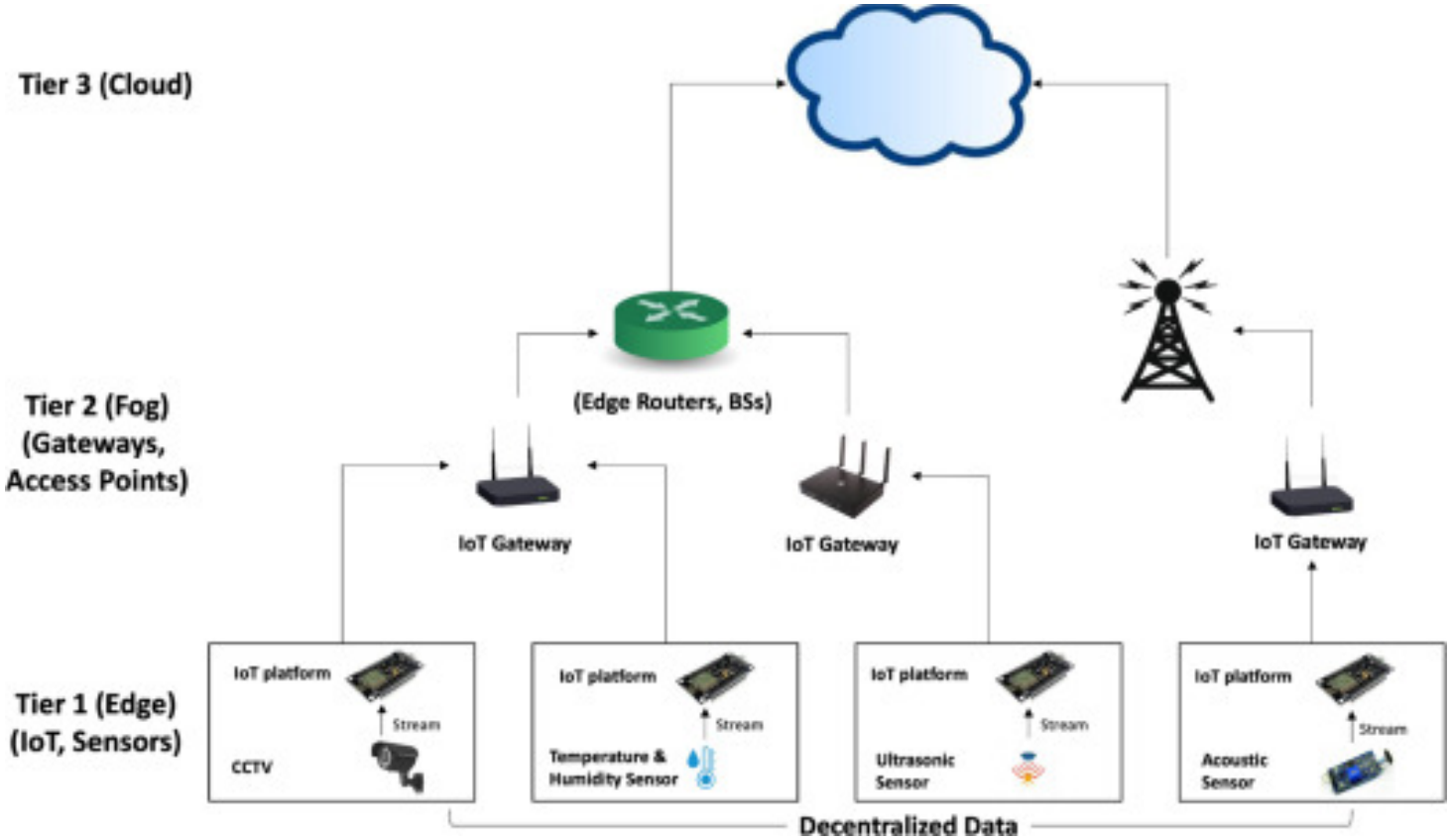


https://www.youtube.com/watch?v=7mYNaVNoIg0

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Challenge: Node-to-Node Communications

- The heart of an IoT system is the ability for Nodes to send messages to one another.
- Each Device or Computer that participates in the IoT system is called a Node.
  - Example: ESP32, Raspberry Pi, Laptop/PC, Virtual Machine on EC2 in AWS, smartphone, etc.
- Each Node must be connected to the other via some form of Network. This can be visualized in the form of a multi-tiered Network Diagram (part of Week 1 topic).



Use MQTT to facilitate this messaging system so your ESP32 sends and receives messages to your PC as part of your Project.

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Takeaway for this Week

1. The MQTT Broker is the heart of most IoT system in the world – allowing IoT devices to send messages to each other across the network.

2. Mosquitto is a popular, open source MQTT Broker, but there are a lot of alternatives such as HiveMQ – an online broker service, that might be easier to try with.

3. There's a large number of tutorials concerning "ESP32 with MQTT and Arduino" so search online to see which ones work best for you. E.g.:
   - MQTT Tutorial using Arduino Framework (donskytech.com)
   - HOW TO CONTROL LED WITH PUBLISH/SUBSCRIBE MQTT IOT (youtube.com)
   - ESP32 MQTT Publish Subscribe with Arduino IDE | Random Nerd Tutorials
   - MQTT on ESP32: A Beginner's Guide | EMQ (emqx.com)

Other PCs can also participate as a MQTT client. Again, look online for similar tutorials:

- Implementing MQTT in Python (hivemq.com)
- How to Use MQTT in C# with MQTTnet | EMQ (emqx.com)

Phone Apps are also available to quickly test MQTT messaging:
Android and MQTT: A Simple Guide. How to develop an MQTT Client with… | by Leonardo Cavagnis | The Startup | Medium

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